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KNOWLEDGE IS NOT STATIC: ORDER-AWARE HYPERGRAPH RAG FOR LANGUAGE MODELS

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ABSTRACT

Retrieval-augmented generation (RAG) enhances large language models by grounding outputs in retrieved knowledge. However, existing RAG methods including graph- and hypergraph-based approaches treat retrieved evidence as an unordered set, implicitly assuming permutation invariance. This assumption is misaligned with many real-world reasoning tasks, where outcomes depend not only on which interactions occur, but also on the order in which they unfold. We propose *Order-Aware Knowledge Hypergraph RAG (OKH-RAG)*, which treats order as a first-class structural property. OKH-RAG represents knowledge as higher-order interactions within a hypergraph augmented with precedence structure, and reformulates retrieval as sequence inference over hyperedges. Instead of selecting independent facts, it recovers coherent interaction trajectories that reflect underlying reasoning processes. A learned transition model infers precedence directly from data without requiring explicit temporal supervision. We evaluate OKH-RAG on order-sensitive question answering and explanation tasks, including tropical cyclone and port operation scenarios. OKH-RAG consistently outperforms permutation-invariant baselines, and ablations show that these gains arise specifically from modeling interaction order. These results highlight a key limitation of set-based retrieval: effective reasoning requires not only retrieving relevant evidence, but organizing it into structured sequences.

***K*eywords** Knowledge Graph · Retrieval-Augmented Generation · Hypergraph Representation · Graph-based RAG

1 Introduction

Large language models (LLMs) have demonstrated impressive capabilities in reasoning and generation, yet their performance on knowledge-intensive tasks remains constrained by the lack of explicit mechanisms for structured inference [1, 2, 3, 4]. Retrieval-augmented generation (RAG) mitigates this limitation by conditioning generation on externally retrieved knowledge, thereby separating knowledge access from language modeling [5, 6, 7, 8]. While early RAG approaches rely on unstructured text retrieval, recent work has shown that incorporating *structured knowledge representations* can significantly improve reasoning fidelity and factual consistency [9, 10].

Graph-based RAG methods represent an important step in this direction by organizing knowledge as a graph of entities and relations [11, 12, 13]. However, conventional graphs encode only *binary* relationships, restricting their ability to model interactions that inherently involve more than two entities. Many phenomena in real-world systems exhibit such higher-order dependencies, where outcomes emerge from the joint configuration of multiple factors rather than pairwise interactions alone. Hypergraphs naturally generalize graphs by allowing edges to connect arbitrary-sized sets of nodes, and therefore provide a principled representation for capturing *higher-order graph dynamics* [14, 15, 16, 17, 18].

Recent work on hypergraph-based RAG demonstrates that representing knowledge with n-ary relations can reduce information loss and improve retrieval effectiveness compared to binary graph formulations [19, 20, 21, 22]. Despite this

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知识不是静态的:面向顺序的超图RAG用于语言模型

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摘要

检索增强生成(RAG)通过将输出与检索到的知识联系起来来增强大型语言模型。然而,现有的RAG 方法,包括基于图和超图的方法,都将检索到的证据视为无序集合,隐含地假设排列不变性。此假设与许多现实世界的推理任务不一致,因为结果不仅取决于哪些交互发生,还取决于它们展开的顺序。我们提出 *Order-Aware Knowledge Hypergraph RAG (OKH-RAG)*,将顺序视为一等的结构属性。OKH-RAG 将知识表示为在带有优先结构的超图中的高阶交互,并将检索重新表述为超边上的序列推理。它不是选择独立的事实,而是恢复反映底层推理过程的连贯交互轨迹。一个学习到的转移模型直接从数据中推断优先级,无需显式时间监督。我们在顺序敏感的任务上评估 OKH-RAG.

K关键词 知识图谱 · 增强检索生成 · 超图表示 · 基于图的RAG

1 引言

大型语言模型(LLMs)在推理和生成方面表现出令人印象深刻的能力,但它们在知识密集型任务上的表现仍然受到缺乏用于结构化推理的明确机制的限制[1, 2, 3, 4]。检索增强生成(RAG)通过基于外部检索的知识进行生成,从而将知识访问与语言建模分离,缓解了这一限制[5, 6, 7, 8]。虽然早期的RAG方法依赖于非结构化文本检索,但近期研究表明,融合*structured knowledge representations*可以显著提高推理的准确性和事实一致性[9, 10]。

基于图的RAG方法通过将知识组织为实体和关系的图[11, 12, 13],在这一方向上代表了重要的一步。然而,传统图只编码*binary*关系,限制了它们对本质上涉及两个以上实体的交互的建模能力。现实世界系统中的许多现象展示了这种高阶依赖,其中结果是由多个因素的联合配置而非仅仅两两交互产生的。超图自然地推广了图的概念,通过允许边连接任意大小的节点集合,因此为捕捉*higher-order graph dynamics*[14, 15, 16, 17, 18]提供了一个有原则的表示。

最近关于基于超图的RAG的研究表明,与二元图表述相比,用n元关系表示知识可以减少信息丢失并提高检索效果 [19, 20, 21, 22]。尽管如此

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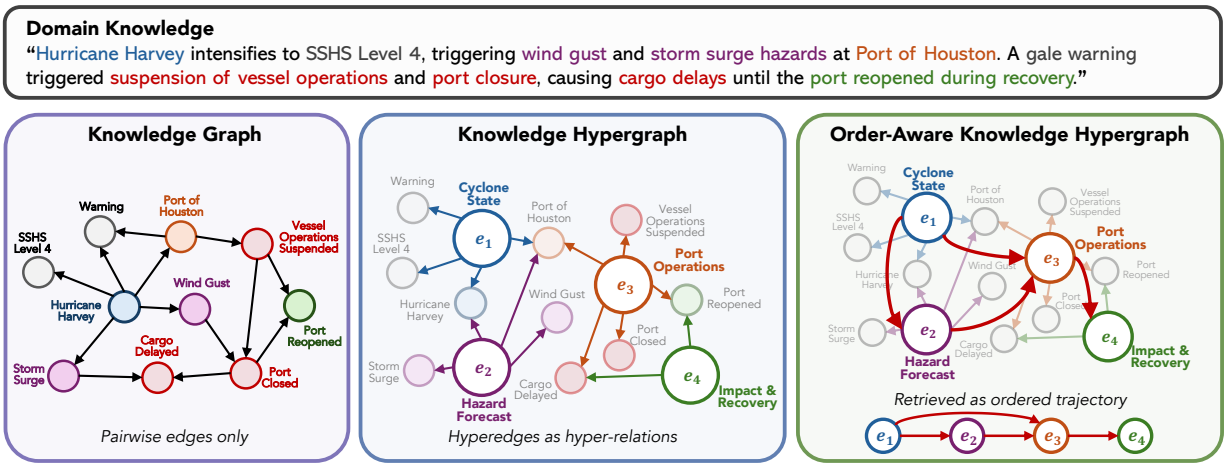


Figure 1: Knowledge graphs fragment multi-entity interactions into pairwise edges; knowledge hypergraphs preserve higher-order relations but retrieve them as unordered sets; our order-aware knowledge hypergraph augments hyperedges with learned precedence, enabling retrieval as ordered trajectories that reflect how interactions unfold.

advance, existing hypergraph-based RAG methods remain fundamentally *static*: hyperedges encode timeless relational facts, and retrieval is performed without accounting for how interactions evolve over time. This static abstraction limits the ability of retrieval mechanisms to support reasoning about processes, causality, and delayed effects.

Though many interaction-aware RAG methods have been proposed, sequential structure is fundamental to knowledge-intensive reasoning, where system-level outcomes typically emerge from ordered interactions rather than simple co-occurrences. Whether representing a chain of subsystem failures or the propagation of causal dependencies, modeling these phenomena as static relations obscures the inherent ordering, persistence, and dynamics of the system, preventing retrieval from capturing underlying dynamics. For example, in a tropical cyclone scenario, port disruption may depend jointly on forecast storm intensity, local infrastructure exposure, evolving port conditions, and operational response. This requires a hypergraph representation, since the relevant dependency is higher-order rather than pairwise. It also requires order-awareness, since the reasoning depends on how these factors evolve over time, from forecast updates to warning escalation, port restrictions, and ultimately operational disruption [23, 24].

In this work, we argue that *order is a fundamental dimension of knowledge structure* in retrieval-augmented generation. We introduce **Order-Aware Knowledge Hypergraph RAG (OKH-RAG)**, as illustrated in Figure 1, a framework that models knowledge as higher-order interactions with explicit precedence and reformulates retrieval as *trajectory inference*. Instead of retrieving a set of independent facts, OKH-RAG retrieves *ordered hyperedge trajectories* that capture how interactions evolve and connect to form coherent reasoning chains. This shifts retrieval from selecting relevant evidence to recovering structured sequences that reflect underlying processes. Our formulation unifies two complementary advances: hypergraph representations for modeling higher-order dependencies, and order-aware inference for capturing interaction dynamics. By integrating these into a single framework, OKH-RAG enables LLMs to reason over processes rather than static snapshots. Our contributions are summarized as follows:

- We introduce an *order-aware knowledge hypergraph* representation that captures higher-order interactions together with their precedence structure.
- We reformulate retrieval as *trajectory inference* over hyperedges, moving from permutation-invariant set retrieval to order-aware sequence retrieval.
- We integrate order-aware hypergraph retrieval into the RAG framework, enabling generation grounded in structured, temporally coherent evidence.

By integrating higher-order structure with interaction order, OKH-RAG establishes a retrieval framework where the *sequence* of evidence, not merely its content, is explicitly modeled, optimized, and empirically validated as a key driver of generation quality.

2 Related Work

Retrieval-Augmented Generation. Retrieval-augmented generation (RAG) enhances parametric language models by conditioning generation on externally retrieved knowledge, mitigating issues such as hallucination, outdated information,

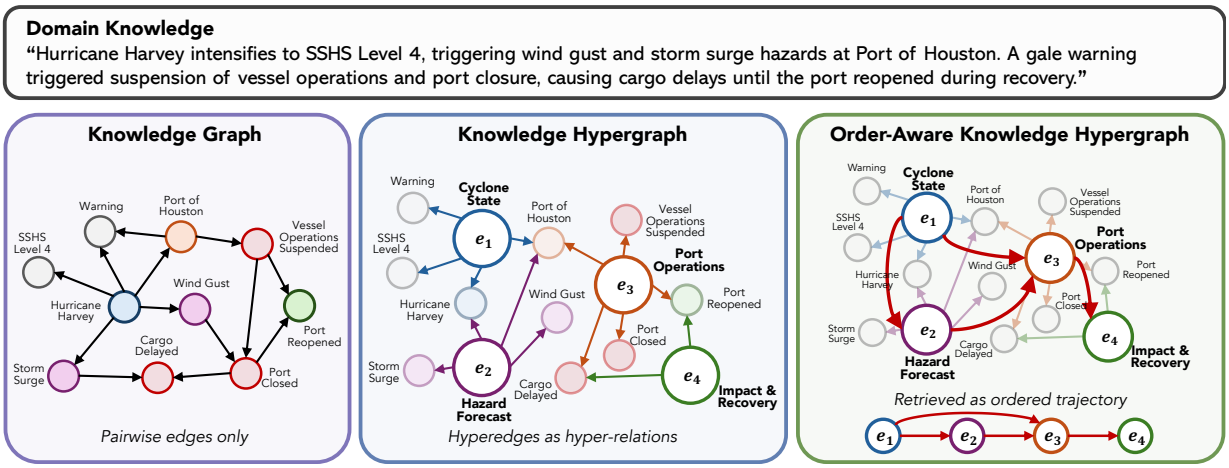


图1:知识图谱将多实体交互分解为成对边;知识超图保留高阶关系,但以无序集合的形式进行检索;我们的顺序感知知识超图通过学习的优先级增强超边,实现按顺序轨迹检索,从而反映交互的展开方式。

尽管已有进展,现有基于超图的 RAG 方法在本质上仍然是 *static*:超边编码的是永恒的关系事实,并且检索是在不考虑交互如何随时间演变的情况下进行的。这种静态抽象限制了检索机制支持关于过程、因果关系和延迟效应的推理的能力。

尽管已经提出了许多考虑交互的RAG方法,但顺序结构对于知识密集型推理是基础的,其中系统级结果通常源于有序的交互,而不是简单的共现。无论是表示子系统故障链还是因果依赖的传播,将这些现象建模为静态关系都会掩盖系统的固有顺序、持久性和动态性,从而导致检索无法捕捉底层动态。例如,在热带气旋情景中,港口中断可能同时依赖于预测风暴强度、当地基础设施暴露情况、不断变化的港口状况以及操作响应。这需要超图表示,因为相关依赖是高阶的而非成对的。它还需要顺序感知,因为推理依赖于这些因素随时间的演变,从预测更新到警告升级、港口限制,最终到操作。

在这项工作中,我们论证了在检索增强生成中的 *order is a fundamental dimension of knowledge structure*。我们提出了有序感知知识超图 RAG(OKH-RAG),如图 1 所示,这是一个将知识建模为具有明确优先级的高阶交互,并将检索重新表述为 *trajectory inference* 的框架。OKH-RAG 并不是检索一组独立的事实,而是检索 *ordered hyperedge trajectories*,以捕捉交互如何演变并连接形成连贯的推理链。这将检索从选择相关证据转变为恢复反映潜在过程的结构化序列。我们的表述统一了两项互补的进展:用于建模高阶依赖关系的超图表示,以及用于捕捉交互动态的有序感知推理。通过将这些整合到单一框架中,OKH-RAG 使大型语言模型能够对过程而非静态快照进行推理。我们的贡献总结如下:

- 我们引入了一种 *order-aware knowledge hypergraph* 表示法,它能够捕捉高阶交互及其优先结构。
- 我们将检索重新表述为对超边的*trajectory inference*,从置换不变的集合检索转向顺序感知的序列检索。
- 我们将顺序感知的超图检索整合到RAG框架中,从而实现基于结构化、时间连贯证据的生成。

通过将高阶结构与交互顺序相结合,OKH-RAG 建立了一个检索框架,其中不仅仅是证据的内容,而是证据的 *sequence* 被明确建模、优化并通过实证验证为生成质量的关键驱动因素。

2 相关工作

检索增强生成。检索增强生成(RAG)通过将生成过程与外部检索的知识相结合来增强参数化语言模型,从而缓解诸如虚构内容、信息过时等问题,

and lack of traceable reasoning [25, 26]. By introducing a retriever that accesses a non-parametric memory, RAG enables models to incorporate domain-specific and up-to-date information at inference time. This framework has been widely adopted in knowledge-intensive applications, particularly in domain-specific settings. Representative applications include biomedical question answering [27], legal document analysis [28], and enterprise knowledge retrieval [29]. Early RAG systems primarily rely on retrieving unstructured text passages using dense retrieval methods such as Dense Passage Retrieval (DPR) [30, 31]. Subsequent surveys and benchmarks [32, 33] identify bottlenecks in both retrieval quality and scalability, particularly with the emergence of large language models (LLMs). Recent work has explored several directions to address these challenges, including knowledge-grounded dialogue systems [34], adaptive retrieval strategies [35, 36, 37], and multi-hop reasoning frameworks [38, 39]. These developments collectively motivate the incorporation of structured representations into the retrieval process.

Graph and Hypergraph-Based RAG. Graph-based RAG methods [40, 41, 42], which organize knowledge as entities and binary relations, represent one of the most effective extensions beyond unstructured retrieval in RAG. These approaches enable graph traversal [43, 44], path-based reasoning [45, 46], and community-level summarization [11], thereby improving evidence aggregation and interpretability compared to purely similarity-based retrieval. However, they remain fundamentally limited by binary representations that decompose higher-order dependencies into pairwise edges, potentially leading to information loss and fragmented reasoning. To address this limitation, hypergraph-based approaches have been introduced to connect arbitrary-sized sets of nodes, naturally capturing n-ary relational facts [47] and higher-order interactions [20, 48]. Such representations provide a more expressive framework for modeling complex relationships and have shown improvements in retrieval efficiency and generation quality over binary graph structures [49]. Nevertheless, existing hypergraph-based RAG methods typically treat hyperedges as *static* facts, failing to capture the dynamics, temporal evolution, and ordering of higher-order interactions, which limits their ability to support more complex reasoning processes.

Temporal Dynamics and Event-Centric Modeling. Temporal and dynamic graph models have been extensively studied to capture time-varying relations and node attributes [50, 51, 52, 53]. These approaches typically focus on predictive tasks such as link forecasting or anomaly detection using message-passing or memory modules [54, 55]. While powerful, these models are largely designed for binary relational structures and do not directly capture higher-order interactions. Furthermore, while recent extensions [56] have introduced temporal hypergraphs to model evolving n-ary relations, they focus on representation learning rather than retrieval-augmented generation. Related work in event-centric modeling emphasizes temporal ordering and causal structure [57, 58], yet these systems often rely on task-specific schemas rather than general-purpose retrieval. OKH-RAG bridges these gaps by unifying hypergraph-based representation with order-awareness. Unlike graph-based RAG, OKH-RAG models higher-order dependencies directly via hyperedges. Unlike static hypergraph methods, it treats order as a first-class dimension, moving from set-based selection to *trajectory-based retrieval*. By treating retrieval as inference over interaction sequences, OKH-RAG enables LLMs to reason about processes and propagation—capabilities previously absent in structured RAG frameworks.

3 Methodology

OKH-RAG is built on the premise that the order of knowledge interactions is a structural property essential for faithful reasoning. While existing RAG systems treat retrieved evidence as an unordered set, we model knowledge as an evolving process and reformulate retrieval as trajectory inference. The framework comprises three stages (Figure 2): (1) constructing an order-aware hypergraph with learned precedence (§ 3.1); (2) retrieving ordered hyperedge trajectories (§ 3.2); and (3) generating responses conditioned on structured evidence chains (§ 3.3).

3.1 Knowledge Hypergraph Construction and Order Learning

The first stage transforms unstructured text into a structured, order-aware representation. Our goal is not only to capture *what* interactions exist, but also *how* they unfold. We first construct a knowledge hypergraph to model higher-order interactions, and then augment it with learned precedence to encode ordering.

Knowledge hypergraph. Many real-world phenomena involve interactions that are inherently higher-order: outcomes depend on the joint configuration of multiple factors rather than pairwise associations. Standard knowledge graphs, restricted to binary edges, cannot represent such interactions without fragmentation. We therefore represent domain knowledge as a *knowledge hypergraph* $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is a set of entities and each hyperedge $e \in \mathcal{E}$ connects an arbitrary subset of entities ($|V_e| \geq 2$). Following **(author?)** [20], we employ *n-ary relational extraction* with natural language descriptions, preserving richer semantics than structured triples. While hypergraphs improve expressiveness in representing interactions, existing hypergraph-based RAG methods treat $\mathcal{H}$ as *static and unordered*: retrieved

以及缺乏可追踪的推理 [25, 26]。通过引入访问非参数记忆的检索器,RAG 使模型能够在推理时整合特定领域的最新信息。该框架已被广泛应用于知识密集型应用,尤其是在特定领域环境中。典型应用包括生物医学问答 [27]、法律文档分析 [28] 和企业知识检索 [29]。早期的 RAG 系统主要依赖于使用密集检索方法(如 Dense Passage Retrieval,DPR)[30, 31] 检索非结构化文本段落。随后的调查和基准测试 [32, 33] 指出检索质量和可扩展性方面的瓶颈,特别是在大语言模型(LLM)出现后。近期研究探索了几种方向来应对这些挑战,包括基于知识的对话系统 [34]、自适应检索策略 [35, 36, 37] 和多跳推理框架 [38, 39]。

基于图和超图的RAG。基于图的RAG方法 [40, 41, 42] 将知识组织为实体和二元关系,是RAG中超越非结构化检索的最有效扩展之一。这些方法支持图遍历 [43, 44]、基于路径的推理 [45, 46] 以及社区级摘要 [11],相比纯基于相似度的检索,提高了证据聚合和可解释性。然而,它们在本质上仍受制于二元表示,这会将高阶依赖分解为两两边,从而可能导致信息丢失和推理碎片化。为了解决这一限制,引入了超图方法,将任意大小的节点集合连接起来,自然地捕捉 n-元关系事实 [47] 和高阶交互 [20, 48]。这种表示提供了一个更具表现力的框架来建模复杂关系,并在检索效率和生成方面表现出改进

时间动态与事件中心建模。时间和动态图模型已被广泛研究,以捕捉随时间变化的关系和节点属性 [50, 51, 52, 53]。这些方法通常侧重于预测任务,如链接预测或异常检测,使用消息传递或记忆模块 [54, 55]。虽然功能强大,但这些模型主要针对二元关系结构设计,并不能直接捕捉高阶交互。此外,尽管近期的扩展[56]引入了时间超图以建模演化的n-元关系,但它们关注的是表示学习而非检索增强生成。事件中心建模的相关工作强调时间顺序和因果结构 [57, 58],然而这些系统往往依赖于特定任务的模式,而非通用的检索。OKH-RAG通过统一基于超图的表示与顺序感知弥合了这些差距。与基于图的RAG不同,OKH-RAG建模了更高的

3 方法论

OKH-RAG 建立在这样的前提上:知识交互的顺序是一种结构性属性,对忠实推理至关重要。现有的 RAG 系统将检索到的证据视为无序集合,而我们将知识建模为一种演变过程,并将检索重新表述为轨迹推断。该框架包括三个阶段(图 2):(1) 构建具有学习到的优先级的顺序感知超图 (§ 3.1);(2) 检索有序超边轨迹 (§ 3.2);以及 (3) 基于结构化证据链生成响应 (§ 3.3)。

3.1 知识超图构建与顺序学习

第一阶段将非结构化文本转换为结构化的、具有顺序感的表示。我们的目标不仅是捕捉存在的*what*交互,还要捕捉它们如何*how*展开。我们首先构建知识超图以建模高阶交互,然后通过学习到的先后关系对其进行增强以编码顺序。

知识超图。许多现实世界中的现象涉及本质上为高阶的相互作用:结果依赖于多个因素的联合配置,而不是成对的关联。受限于二元边的标准知识图,无法在不破碎的情况下表示这种相互作用。因此,我们将领域知识表示为 *knowledge hypergraph* $\mathcal{H} = (\mathcal{V}, \mathcal{E})$,其中 $\mathcal{V}$ 是一组实体,每条超边 $e \in \mathcal{E}$ 连接任意子集的实体($|V_e| \geq 2$)。遵循 (author?) [20],我们使用 *n-ary relational extraction* 及其自然语言描述,保留比结构化三元组更丰富的语义。虽然超图在表示相互作用方面提高了表达能力,但现有基于超图的 RAG 方法将 $\mathcal{H}$ 视为 *static and unordered*:检索

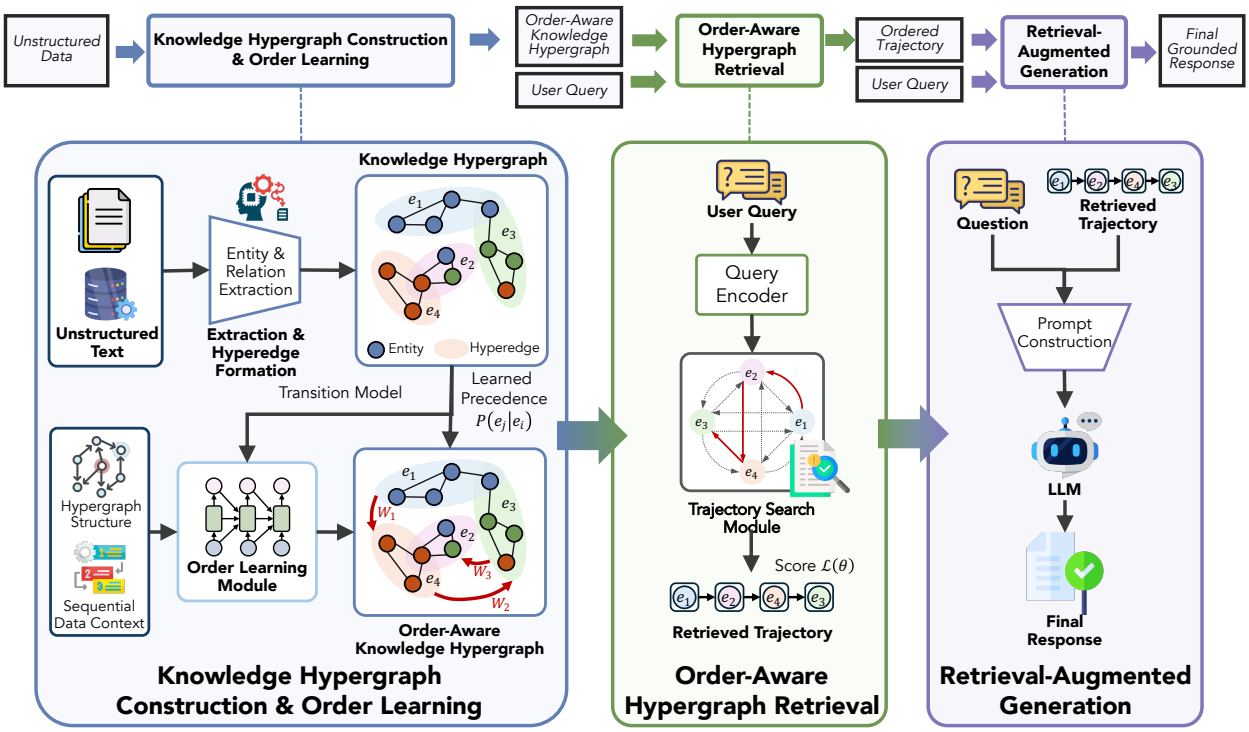


Figure 2: **Overview of OKH-RAG.** The framework constructs an order-aware knowledge hypergraph from documents, retrieves query-specific interaction trajectories via sequence inference, and generates responses conditioned on structured, temporally coherent evidence.

hyperedges are presented as a permutation-invariant set. As we show below, this abstraction is insufficient for reasoning tasks in which order affects inference.

Entities. Each entity $v \in \mathcal{V}$ represents a distinct object, concept, or state, defined as $v = (n_v,\ \tau_v,\ d_v,\ c_v)$, where n_v is the entity name, $\tau_v \in \mathcal{T}$ is a type from a domain-specific type system, d_v is a natural-language explanation, and $c_v \in (0, 1]$ is a confidence score. The type system distinguishes *persistent objects* (e.g., a port or cyclone) that anchor the hypergraph, *transient states* (e.g., a cyclone state at a specific horizon) that capture evolution, and *temporal anchors* (e.g., `horizon:T-48`) that index interactions to positions in the ordering.

Hyperedges. Entities interact through hyperedges—the primary knowledge-carrying units. Each hyperedge is a tuple $e = (V_e,\ r_e,\ s_e,\ \mathbf{a}_e,\ c_e)$, where $V_e \subseteq \mathcal{V}$ ($|V_e| \geq 2$) is the participating entity set, $r_e \in \mathcal{R}$ is a typed relation from a controlled vocabulary, s_e is a *natural language description* capturing the interaction's semantic content, $\mathbf{a}_e$ is a set of key–value attributes recording quantitative properties, and $c_e \in (0, 1]$ is a confidence score. The natural language description s_e is critical: unlike structured triples, it can express multi-entity dependencies whose meaning arises from joint configuration rather than pairwise association. The vocabulary is defined in Appendix A.1.

N-ary relational extraction. We construct $\mathcal{H}$ from a document corpus $\mathcal{K}$ using a language model π guided by an extraction prompt p_{ext}: $\mathcal{F}^d = \{f_1, \ldots, f_k\} \sim \pi(\mathcal{F} \mid p_{\text{ext}},\ d)$, where each fact $f_i = (e_i, V_{e_i})$ pairs a hyperedge with its entity set. For multi-horizon documents, per-horizon extraction followed by merging yields higher coverage than monolithic processing. Post-extraction normalization canonicalizes identifiers, injects horizon entities, and synthesizes cross-horizon edges (Appendix A.2). The complete hypergraph aggregates across documents:

$$\mathcal{H} = \Big(\bigcup_{d \in \mathcal{K}} \bigcup_{f_i \in \mathcal{F}^d} V_{e_i},\ \bigcup_{d \in \mathcal{K}} \{e_i \mid f_i \in \mathcal{F}^d\} \Big). \tag{1}$$

Order-aware hypergraph. The knowledge hypergraph captures *what* interactions exist but not their order. To encode precedence while remaining agnostic to clock time, we introduce a discrete sequence index $\ell \in \{1, \ldots, L\}$ capturing

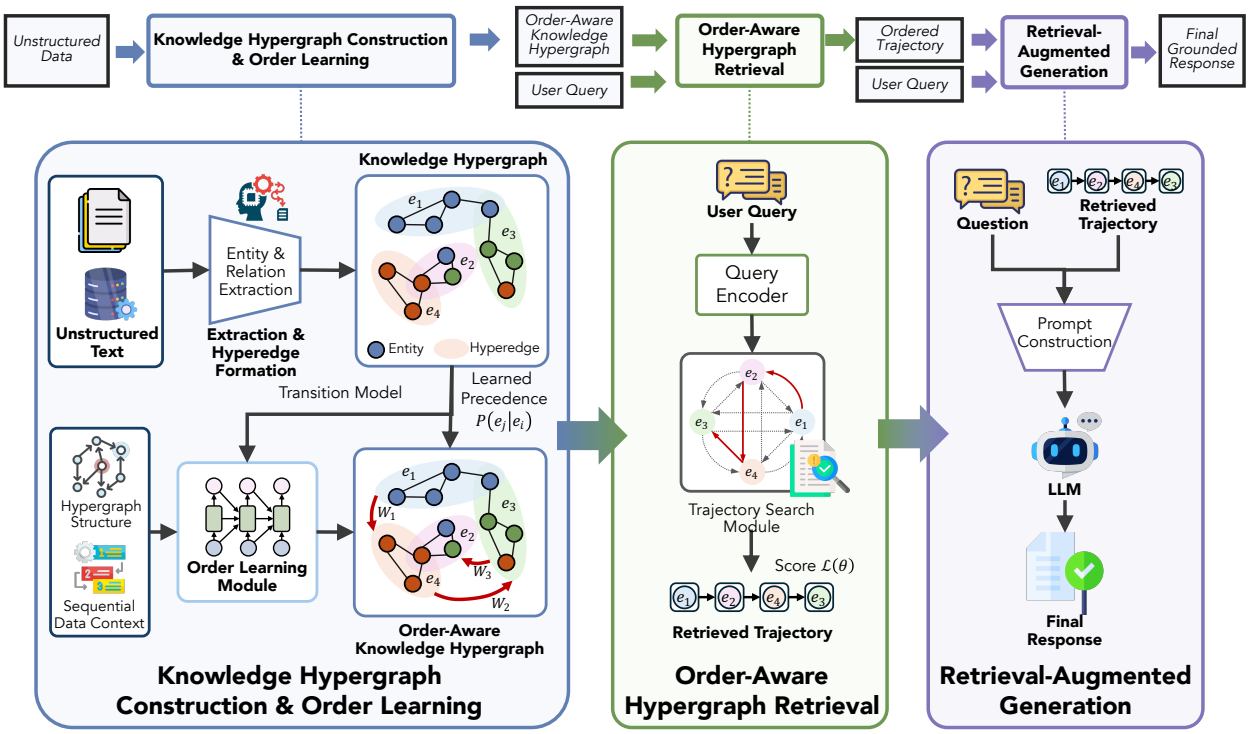


图 2:OKH-RAG 概述。该框架从文档中构建有序感知的知识超图,通过序列推理检索特定查询的交互轨迹,并基于结构化、时间上连贯的证据生成响应。

超边被表示为一个排列不变的集合。正如我们下面所展示的,这种抽象对于顺序影响推理的推理任务是不充分的。

实体。每个实体 $v \in \mathcal{V}$ 代表一个不同的对象、概念或状态,定义为 $v = (n_v,\ \tau_v,\ d_v,\ c_v)$,其中 n_v 是实体名称,$\tau_v \in \mathcal{T}$ 是来自特定领域类型系统的类型,d_v 是自然语言解释,$c_v \in (0,\ 1]$ 是置信度分数。类型系统区分 *persistent objects* (例如,一个端口或气旋),用于锚定超图,*transient states* (例如,在特定时间点的气旋状态),用于捕捉演变,以及 *temporal anchors* (例如,horizon:T-48),用于将交互索引到顺序中的位置。

超边。实体通过超边进行交互——这是主要的知识承载单元。每个超边是一个元组 $e = (V_e,\ r_e,\ s_e,\ \mathbf{a}_e,\ c_e)$,其中 $V_e \subseteq \mathcal{V}$ ($|V_e| \geq \in$) 是参与的实体集合,$r_e \in \mathcal{R}$ 是来自受控词汇的类型化关系,s_e 是一个 *natural language description* 捕捉交互语义内容,$\mathbf{a}_e$ 是记录定量属性的键值属性集合,$c_e \in$ ($\in$ [0,1] 是置信度分数。自然语言描述 s_e 至关重要:与结构化三元组不同,它可以表达多实体依赖,其意义源自联合配置而非成对关联。词汇表在附录 A.1 中定义。

N-ary 关系抽取。我们使用语言模型 π 并依据抽取提示 p_{ext} 从文档语料库 $\mathcal{K}$ 构建 $\mathcal{H}$: $\mathcal{F}^d = \{f_1, \ldots, f_k\} \sim \pi(\mathcal{F} \mid p_{\text{ext}},\ d)$,其中每个事实 $f_i = (e_i, V_{e_i})$ 将一个超边与其实体集合配对。对于多视角文档,先按视角抽取再合并,比整体处理能获得更高的覆盖率。抽取后规范化会规范化标识符、注入视角实体,并综合生成跨视角边(附录 A.2))。完整超图在文档之间进行聚合:

$$\mathcal{H} = \Big(\bigcup_{d \in \mathcal{K}} \bigcup_{f_i \in \mathcal{F}^d} V_{e_i},\ \bigcup_{d \in \mathcal{K}} \{e_i \mid f_i \in \mathcal{F}^d\} \Big). \tag{1}$$

顺序感知超图。知识超图捕捉 *what* 交互的存在,但不包括它们的顺序。为了在保持对时钟时间不可知的情况下编码优先顺序,我们引入了离散序列索引 $\ell \in \{1, \ldots, L\}$ 来捕捉

relative order and represent knowledge as a sequence of states $\mathcal{H}^{(\ell)} = (\mathcal{V}, \mathcal{E}^{(\ell)})$. This induces a precedence relation $e_i \prec e_j \Leftrightarrow \exists\, \ell_1 < \ell_2$ s.t. $e_i \in \mathcal{E}^{(\ell_1)}, e_j \in \mathcal{E}^{(\ell_2)}$, yielding the *order-aware knowledge hypergraph*:

$$\mathcal{H}_{\prec} = (\mathcal{V},\ \mathcal{E},\ \prec). \tag{2}$$

The index ℓ encodes precedence, not duration; $\prec$ is a partial order accommodating concurrent interactions; and when $\prec = \emptyset$, the representation reduces to a standard unordered hypergraph. A non-empty $\prec$ breaks permutation invariance and enables order-aware retrieval. The precedence graph is constructed via domain-informed structural rules (Appendix A.3).

Learning precedence. Since explicit precedence annotations are rarely available, we learn a parametric transition model:

$$P_\theta(e_j \mid e_i) = \frac{\exp(\mathbf{h}_{e_i}^\top W\, \mathbf{h}_{e_j})}{\sum_{e'} \exp(\mathbf{h}_{e_i}^\top W\, \mathbf{h}_{e'})}, \tag{3}$$

where $\mathbf{h}_e \in \mathbb{R}^d$ is a dense hyperedge embedding and W is a learnable weight matrix (low-rank factorized as $U^\top V$, $r \ll d$). The bilinear form is inherently asymmetric—$P_\theta(e_j \mid e_i) \neq P_\theta(e_i \mid e_j)$—encoding directionality without explicit temporal features.

The model is trained via a contrastive objective with three self-supervised signals: *document order* (positional adjacency as a proxy for precedence), *entity-overlap consistency* (shared entities as evidence for co-participation in reasoning chains), and *retrieval-induced preference* (reinforcing transitions along empirically successful trajectories in a self-training loop):

$$\mathcal{L}(\theta) = \mathbb{E}_{(e_i,e_j)\sim\mathcal{P}}[-\log P_\theta(e_j \mid e_i)] + \alpha\, \mathbb{E}_{(e_i,e_m)\sim\mathcal{N}}[\log P_\theta(e_m \mid e_i)], \tag{4}$$

where $\mathcal{P}$ and $\mathcal{N}$ are positive and negative ordered pairs (details in Appendix A.4).

Order sensitivity of knowledge. We have described how to construct and learn $\mathcal{H}_{\prec}$. Is this machinery necessary? We establish formally that the answer is yes.

Proposition 1 (Order sensitivity)**.** *There exist k_a, k_b, q such that $P(y \mid k_a, k_b, q) \neq P(y \mid k_b, k_a, q)$. Any permutation-invariant retrieval method is therefore insufficient for modeling $P(y \mid q)$ when reasoning depends on interaction order.*

The proposition requires no assumptions about timestamps or causality—only that evidence order can alter inferred outcomes. This complements **(author?)** [20]: where they establish that hypergraphs are more expressive in *what* they represent, we establish that order-aware hypergraphs are more expressive in *how* they can be retrieved (discussion in Appendix A.5).

3.2 Order-aware Hypergraph Retrieval

Standard retrieval scores each element independently and returns the highest-scoring set, treating evidence selection as a ranking problem. This discards arrangement. For order-sensitive reasoning, the quality of retrieved evidence depends not only on its members but on their sequence. We reformulate retrieval as *sequence inference*: given q, recover the highest-scoring ordered trajectory through $\mathcal{H}_{\prec}$.

Retrieval objective. We seek an ordered trajectory $\gamma = (e^{(1)}, \ldots, e^{(L)})$ maximizing:

$$\gamma^* = \arg\max_{\gamma} \underbrace{\sum_{k=1}^{L} \mathrm{Rel}(e^{(k)}, q)}_{\text{relevance}} + \lambda \underbrace{\sum_{k=1}^{L-1} \log P_\theta(e^{(k+1)} \mid e^{(k)})}_{\text{order coherence}} + \mu \cdot \underbrace{\mathrm{Prec}(\gamma)}_{\substack{\text{precedence}\\ \text{consistency}}} + \nu \cdot \underbrace{\mathrm{Ovlp}(\gamma)}_{\substack{\text{entity}\\ \text{continuity}}} + \rho \cdot \underbrace{\mathrm{Cov}(\gamma)}_{\substack{\text{phase}\\ \text{coverage}}} \tag{5}$$

Relevance ensures topical pertinence via cosine similarity. *Order coherence* ensures consecutive interactions form plausible transitions under P_θ. *Precedence consistency* enforces alignment with the structural relation $\prec$. *Entity continuity* rewards entity sharing between consecutive steps, favoring coherent chains over disjointed fact assemblages. *Phase coverage* rewards spanning distinct reasoning stages (advisory → hazard → operation → impact → recovery), preventing narrow evidence concentration. The hyperparameters $\lambda, \mu, \nu, \rho \geq 0$ are modular: setting all to zero recovers standard top-k retrieval (formal definitions in Appendix A.6).

相对顺序并将知识表示为状态序列 $\mathcal{H}^{(\ell)} = (\mathcal{V}, \mathcal{E}^{(\ell)})$。这引出了一个优先关系 $e_i \prec e_j \Leftrightarrow \exists \ell_1 < \ell_2$,使得 $e_i \in \mathcal{E}^{(\ell_1)}, e_j \in \mathcal{E}^{(\ell_2)}$,产生 *order-aware knowledge hypergraph*:

$$\mathcal{H}_{\prec} = (\mathcal{V},\ \mathcal{E},\ \prec). \tag{2}$$

索引 ℓ 编码优先级,而不是持续时间;$\prec$ 是一个部分顺序,能够容纳并发交互;当 $\prec = \emptyset$ 时,该表示简化为一个标准的无序超图。非空的 $\prec$ 会破坏排列不变性,并使得检索能够考虑顺序。优先图是通过领域信息的结构规则构建的(附录 A.3)。

学习优先级。由于显式的优先级注释很少可用,我们学习一个参数化的转移模型:

$$P_\theta(e_j \mid e_i) = \frac{\exp(\mathbf{h}_{e_i}^\top W\, \mathbf{h}_{e_j})}{\sum_{e'} \exp(\mathbf{h}_{e_i}^\top W\, \mathbf{h}_{e'})}, \tag{3}$$

其中 $\mathbf{h}_e \in \mathbb{R}^d$ 是一个密集超边嵌入,W 是一个可学习的权重矩阵(低秩分解为 $U^\top V$,$r \ll d$)。双线性形式本质上是不对称的—$P_\theta(e_j \mid e_i) \neq P_\theta(e_i \mid e_j)$—编码方向性而无需显式的时间特征。

该模型通过对比目标进行训练,采用三个自监督信号:*document order* (将位置邻近作为前后顺序的代理), *entity-overlap consistency* (以共享实体作为推理链中共同参与的证据),以及*retrieval-induced preference* (在自我训练循环中沿经验上成功的轨迹强化过渡):

$$\mathcal{L}(\theta) = \mathbb{E}_{(e_i,e_j)\sim\mathcal{P}}[-\log P_\theta(e_j \mid e_i)] + \alpha\, \mathbb{E}_{(e_i,e_m)\sim\mathcal{N}}[\log P_\theta(e_m \mid e_i)], \tag{4}$$

在哪里 $\mathcal{P}$ and $\mathcal{N}$ 是正负有序对(详情见附录) dix A.4)。

知识的顺序敏感性。我们已经描述了如何构建和学习 $\mathcal{H}_{\prec}$。这种机制有必要吗?我们正式确定答案是肯定的。

命题1(顺序敏感性)。*There exist k_a, k_b, q such that $P(y \mid k_a, k_b, q) \neq P(y \mid k_b, k_a, q)$. Any permutation-invariant retrieval method is therefore insufficient for modeling $P(y \mid q)$ when reasoning depends on interaction order.*

该命题不需要关于时间戳或因果关系的任何假设——只要求证据的顺序可以改变推断的结果。这补充了(作者?)[20]:在他们确定超图在*what*中表示得更具表达力的地方,我们确定在*how*中可以检索的顺序感知超图更具表达力(讨论见附录A.5)。

3.2 顺序感知超图检索

标准检索对每个元素进行独立评分,并返回得分最高的集合,将证据选择视为排序问题。这会忽略排列顺序。对于顺序敏感的推理,检索到的证据质量不仅取决于其成员,还取决于它们的顺序。我们将检索重新表述为 *sequence inference*:给定 q,在 $\mathcal{H}_{\prec}$ 中恢复得分最高的有序轨迹。

检索目标。我们寻求一个有序轨迹 $\gamma = (e^{(1)}, \ldots, e^{(L)})$ 以最大化:

$$\gamma^* = \arg\max_{\gamma} \underbrace{\sum_{k=1}^{L} \mathrm{Rel}(e^{(k)}, q)}_{\text{relevance}} + \lambda \underbrace{\sum_{k=1}^{L-1} \log P_\theta(e^{(k+1)} \mid e^{(k)})}_{\text{order coherence}} + \mu \cdot \underbrace{\mathrm{Prec}(\gamma)}_{\substack{\text{precedence}\\ \text{consistency}}} + \nu \cdot \underbrace{\mathrm{Ovlp}(\gamma)}_{\substack{\text{entity}\\ \text{continuity}}} + \rho \cdot \underbrace{\mathrm{Cov}(\gamma)}_{\substack{\text{phase}\\ \text{coverage}}} \tag{5}$$

Relevance 通过余弦相似度确保主题相关性。*Order coherence* 确保连续交互在 P_θ 下形成合理的过渡。*Precedence consistency* 强制与结构关系 $\prec$ 对齐。*Entity continuity* 对连续步骤之间的实体共享给予奖励,偏向连贯链而非分散的事实集合。*Phase coverage* 奖励跨越不同推理阶段(建议 $\rightarrow$ 风险 $\rightarrow$ 操作 $\rightarrow$ 影响 $\rightarrow$ 恢复),防止证据集中过窄。超参数 $\lambda, \mu, \nu, \rho \geq 0$ 是模块化的:将所有参数设为零可恢复标准 top-k 检索(正式定义见附录 A.6)。

Candidate scoping and inference. We scope the candidate set $\mathcal{C}$ via top-K cosine retrieval followed by group-aware expansion (entity overlap and domain group membership), yielding $|\mathcal{C}|$. *Beam search* serves as the primary inference algorithm; *Viterbi DP* provides exact optimization for smaller candidate sets (Appendix A.7).

Multi-trajectory retrieval. Many queries admit multiple valid reasoning paths. We therefore retrieve a set of diverse trajectories $\Gamma_q = \{\gamma_1, \ldots, \gamma_N\}$, enabling the model to consider alternative explanations and complementary evidence chains.

3.3 Retrieval-augmented Generation

Concatenating hyperedge texts into a flat context window would collapse ordered trajectories back into unordered sets. OKH-RAG instead presents each trajectory as a numbered evidence chain with explicit structural annotations.

Structured evidence and generation. Each hyperedge $e^{(k)}$ in a trajectory carries a *step index*, a *horizon label*, a *phase label*, and *entity provenance*—making the reasoning structure legible to the generator. The language model produces $y \sim P(y \mid q, \Gamma_q)$, and can thereby recognize preconditions, track escalation across horizons, and trace downstream consequences—capabilities inaccessible to generators receiving unordered evidence. When multiple trajectories are available, the primary mode presents all paths in a single prompt for cross-referencing; a fallback mode aggregates independent per-trajectory answers via confidence-weighted voting or averaging (Appendix A.8).

4 Experiments

We evaluate the central hypothesis of this paper: *when answers depend on how evidence unfolds, retrieval should preserve order rather than treat evidence as an unordered set.* Our experiments address four questions: (1) Does the constructed order-aware knowledge hypergraph capture the structural regimes in which order matters? (2) Does order-aware retrieval recover coherent, query-adaptive evidence trajectories? (3) Does this improve QA performance over permutation-invariant baselines? (4) Do the gains arise specifically from modeling order?

4.1 Experimental Setup

We evaluate on **CyPortQA** [24], a domain-specific QA benchmark for tropical cyclone–port impact assessment. CyPortQA contains 2,917 real-world disruption scenarios from 2015 to 2023, spanning 145 U.S. principal ports and 90 named storms. Each scenario includes multi-horizon descriptions from T-120 to T-12 covering storm evolution, hazard forecasting, operational response, and impact prediction. The benchmark contains 117,178 questions across four types: True/False (TF), Multiple Choice (MC), Short Answer / Numeric (SA), and Text Description (TD). Because many questions require combining evidence across forecast horizons, CyPortQA provides a natural testbed for order-aware retrieval.

Baselines. We compare OKH-RAG against baselines that span the progression from unstructured to structured retrieval. **Text-RAG** performs dense retrieval over unstructured text chunks and concatenates the top-k results as context. **GraphRAG** [11] retrieves over a binary knowledge graph using graph traversal and community summarization. **HyperGraphRAG** [20] retrieves unordered hyperedges from the same knowledge hypergraph $\mathcal{H}$ without modeling order. **OKH-RAG** retrieves order-aware hyperedge trajectories from $\mathcal{H}_\prec$ using the full objective in Equation 5. All methods use the same generator (GPT-4o) and the same embedding model (`text-embedding-3-small`), isolating the effect of retrieval structure. For OKH-RAG, we use beam search with $B = 8$, $L = 8$, $N = 3$, and default hyperparameters $(\lambda, \mu, \nu, \rho) = (1.2, 0.3, 0.2, 0.5)$.

Evaluation metrics. We evaluate answer accuracy across four question types. For True/False (TF) and Multiple Choice (MC), we use exact match; for Short Answer / Numeric (SA), tolerance-based accuracy; and for Text Description (TD), an LLM-judged semantic score in $[0, 1]$ using GPT-4o as the evaluator.

4.2 Structure of the constructed knowledge hypergraph

We first ask whether the extracted knowledge exhibits the kinds of structure that make order-aware retrieval necessary. Figure 3 contrasts two scenarios at opposite ends of a structural spectrum. The hypergraph for Hurricane ALEX (left panel) shows strong *within-horizon regularity*: across horizons, the extracted hyperedges follow nearly identical local phase patterns, yielding clean horizon-stratified clusters. Structural entities such as the cyclone, port, and temporal anchors connect broadly, while transient states remain largely confined to individual horizons. In this regime, unordered

候选范围界定与推理。我们通过前 K 个余弦检索对候选集合 $\mathcal{C}$ 进行范围界定,然后进行考虑群组的扩展(实体重叠和领域群组成员关系),得到 $|\mathcal{C}|$。*Beam search* 作为主要推理算法;*Viterbi DP* 为较小的候选集合提供精确优化(附录 A.7)。

多轨迹检索。许多查询存在多个有效的推理路径。因此,我们检索一组多样化的轨迹 $\Gamma_q = \{\gamma_1, \ldots, \gamma_N\}$,使模型能够考虑替代解释和互补的证据链。

3.3 检索增强生成

将超边文本连接成一个平铺的上下文窗口会将有序轨迹压缩回无序集合。OKH-RAG 则将每条轨迹呈现为带有明确结构注释的编号证据链。

结构化证据与生成。轨迹中的每个超边 $e^{(k)}$ 都携带一个 *step index*、一个 *horizon label*、一个 *phase label* 和 *entity provenance* —— 使推理结构对生成器可读。语言模型生成 $y \sim P(y \mid q, \Gamma_q)$,从而能够识别前提条件、跟踪跨时间跨度的升级,并追踪下游后果 —— 这些能力对于接收无序证据的生成器来说是无法获得的。当有多条轨迹可用时,主要模式在单个提示中呈现所有路径以便交叉参考;备用模式通过基于置信度的投票或平均(附录 A.8)汇总独立的每条轨迹答案。

4 个实验

我们评估本文的核心假设:*when answers depend on how evidence unfolds, retrieval should preserve order rather than treat evidence as an unordered set.* 我们的实验涉及四个问题:(1)所构建的顺序感知知识超图是否捕捉了顺序重要的结构模式?(2)顺序感知检索是否能够恢复连贯的、适应查询的证据轨迹?(3)这是否提升了 QA 性能,相对于排列不变的基线方法?(4)这些提升是否专门源自于顺序建模?

4.1 实验设置

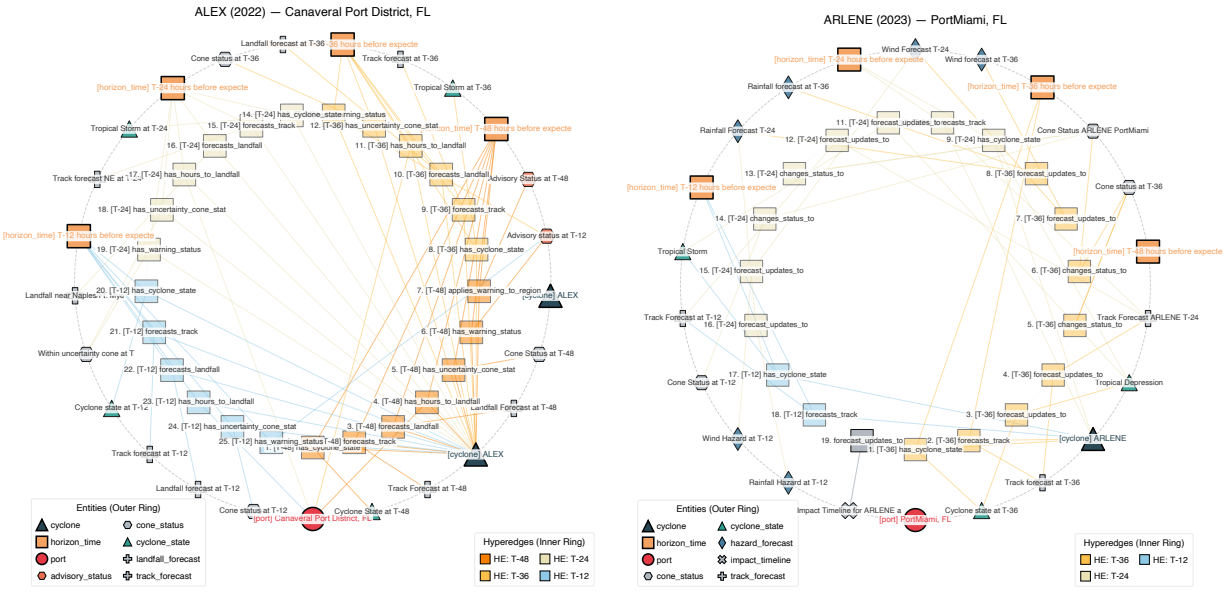
我们在 CyPortQA [24] 上进行评估,这是一个针对热带气旋对港口影响评估的领域特定问答基准。CyPortQA 包含 2015 年至 2023 年的 2,917 个真实中断场景,覆盖 145 个美国主要港口和 90 个命名风暴。每个场景包括从 T-120 到 T-12 的多时段描述,涵盖风暴演变、灾害预测、操作响应和影响预测。该基准包含 117,178 个问题,分为四种类型:判断题(TF)、多项选择题(MC)、简答/数字题(SA)和文本描述题(TD)。由于许多问题需要结合不同预测时段的证据,CyPortQA 提供了一个自然的顺序感知检索测试平台。

基线。我们将 OKH-RAG 与从非结构化检索到结构化检索的基线方法进行比较。Text-RAG 在非结构化文本块上执行密集检索,并将排名前 k 的结果拼接作为上下文。GraphRAG [11] 使用图遍历和社区总结在二元知识图上进行检索。HyperGraphRAG [20] 从相同的知识超图中检索无序超边 $\mathcal{H}$,而不建模顺序。OKH-RAG 使用公式 5 中的完整目标从 $\mathcal{H}_\prec$ 检索考虑顺序的超边轨迹。所有方法均使用相同的生成器(GPT-4o)和相同的嵌入模型(text-embedding-3-small),从而单独考察检索结构的影响。对于 OKH-RAG,我们使用束搜索,$B = 8$、$L = 8$、$N = 3$,以及默认超参数 $(\lambda, \mu, \nu, \rho) = (1.2, 0.3, 0.2, 0.5)$。

评估指标。我们评估四种问题类型的答案准确性。对于是非题(TF)和选择题(MC),我们使用完全匹配;对于简答题/数值题(SA),使用基于容差的准确性;对于文本描述题(TD),使用 GPT-4o 作为评估者的语义评分,范围在 [0, 1] 之间。

4.2 构建的知识超图结构

我们首先问提取出的知识是否展现出那种需要顺序感知检索的结构。图3对比了结构光谱两端的两种情景。飓风ALEX的超图(左图)显示出强烈的*within-horizon regularity*:跨越不同的层位,提取的超边几乎遵循相同的局部相位模式,形成清晰的按层位划分的聚类。诸如气旋、港口和时间锚等结构实体具有广泛的连接,而短暂状态主要局限于各个层位。在这种情况下,无序的



(a) ALEX (2022), Canaveral. (b) ARLENE (2023), PortMiami.

Figure 3: **Two structural regimes in the knowledge hypergraph.** Outer ring shows typed entities; inner ring shows ordered hyperedges.

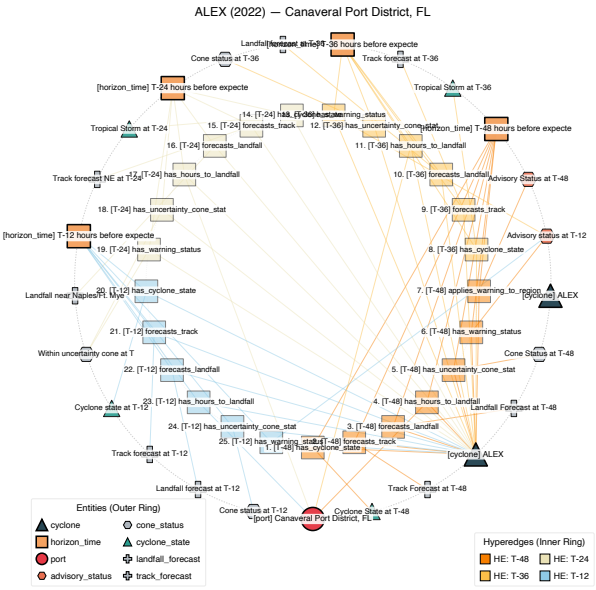
retrieval can often succeed by selecting the correct local snapshot. Tropical Storm ARLENE (right panel) shows a different regime. Nearly half of its hyperedges are *cross-horizon transitions*, such as `forecast_updates_to` and `changes_status_to`, linking states across horizons and introducing dependencies that unfold over time. Where ALEX forms largely separable clusters, ARLENE exhibits visible inter-horizon structure. This contrast highlights when order-awareness matters most. In phase-regular scenarios, retrieval is largely local. In evolution-rich scenarios, the key challenge is not only retrieving the right hyperedges, but retrieving them in the right sequence. The order-aware hypergraph $\mathcal{H}_{\prec}$ is designed to support both regimes within a unified representation.

4.3 QA retrieval results

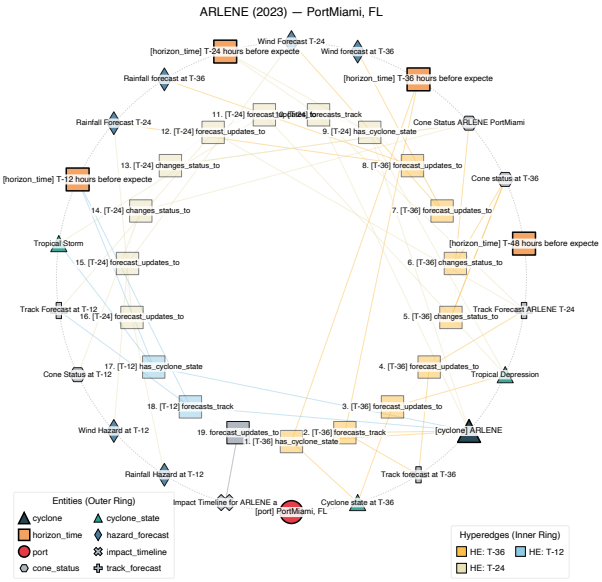
We next examine whether OKH-RAG adapts its retrieved trajectories to the reasoning demands of the query. Figure 4 shows two examples for Hurricane ARTHUR (2020). The first question, "What is the expected landfall location?", requires *cross-horizon reasoning*. The retrieved trajectory begins at T-36 with cyclone state, track, and landfall evidence, then transitions to T-24 for a second round of timing, probability, and landfall information. Rather than collecting isolated facts, the trajectory assembles an evolving forecast chain, allowing the generator to synthesize the answer "Outer Banks, North Carolina" with high confidence. The second question, "Which is Baltimore's closest weather forecast location?", exhibits *within-horizon factual retrieval*. Here the trajectory remains at T-12 and follows a compact local chain centered on the answer entity, which appears early and is reinforced later through hazard-related context. This yields an exact match with high confidence. The contrast is informative: for the landfall question, retrieval prioritizes breadth across horizons; for the station question, it prioritizes depth within a single horizon. In both cases, retrieval recovers an evidence chain aligned with the query's reasoning demands rather than a flat list of relevant hyperedges.

4.4 Comparison with baselines

Table 1 reports answer accuracy across all four question types. The results show a consistent progression from unstructured retrieval to structured, order-aware retrieval. Moving from Text-RAG to GraphRAG yields a large gain, indicating the value of explicit relational structure. Replacing binary graphs with hypergraphs yields a further improvement, confirming the importance of higher-order interactions. OKH-RAG performs best on all four question types and achieves the highest overall accuracy. The gain over HyperGraphRAG is particularly important because both methods use the same underlying hypergraph representation; the difference is whether retrieved evidence is treated as an unordered set or an ordered trajectory. This isolates the contribution of order-awareness beyond the representational



(a) ALEX (2022), Canaveral.



(b) ARLENE (2023), PortMiami.

图3:知识超图中的两种结构模式。外环显示类型化实体;内环显示有序超边。

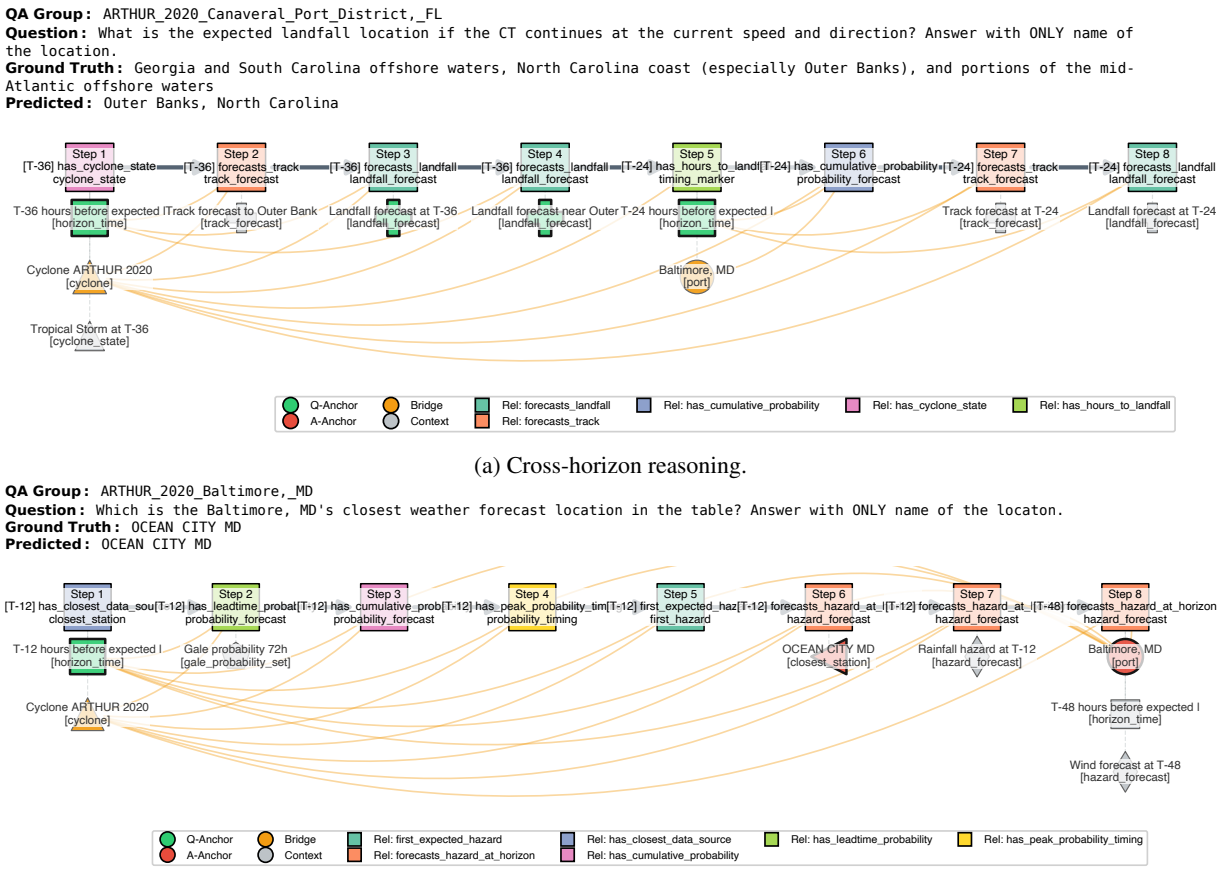
通过选择正确的本地快照,检索通常可以成功。热带风暴 ARLENE(右图)显示了一个不同的模式。它几乎一半的超边是 *cross-horizon transitions*,例如 forecast_updates_to 和 changes_status_to,将各个时间段的状态连接起来,并引入随时间展开的依赖关系。与 ALEX 主要形成可分离的簇不同,ARLENE 展现出明显的跨时间结构。这种对比凸显了顺序意识何时最为重要。在规律性阶段场景中,检索主要是本地的。在演化丰富的场景中,关键挑战不仅在于检索正确的超边,还在于以正确的顺序检索它们。有序超图 $\mathcal{H}_{\prec}$ 的设计旨在在统一表示中支持这两种模式。

4.3 QA 检索结果

接下来,我们考察 OKH-RAG 是否会根据查询的推理需求调整其检索到的轨迹。图 4 展示了 2020 年飓风 ARTHUR 的两个示例。第一个问题,“预计登陆地点在哪里?”,需要 *cross-horizon reasoning*。检索到的轨迹从 T-36 开始,包括气旋状态、路径和登陆证据,然后过渡到 T-24 进行第二轮时间、概率和登陆信息的评估。轨迹不是简单地收集孤立的事实,而是组装出一个不断发展的预测链,使生成器能够以高度自信综合出答案“北卡罗来纳州外滩”。第二个问题,“巴尔的摩最近的天气预报地点是哪一个?”,表现出 *within-horizon factual retrieval*。在这里,轨迹停留在 T-12,沿着围绕答案实体的紧凑本地链展开,该实体出现较早,并通过与危险相关的上下文在后续得到强化。这产生了与答案的高置信度完全匹配。这个对比很有启发性:对于登陆问题,检索

4.4 与基线的比较

表 1 报告了四种问题类型的答案准确率。结果显示,从非结构化检索到结构化、考虑顺序的检索呈现出一致的进步。从 Text-RAG 转向 GraphRAG 带来了显著提升,表明显式关系结构的价值。将二元图替换为超图进一步提高了性能,证实了高阶交互的重要性。OKH-RAG 在四种问题类型上表现最佳,并取得了最高的整体准确率。与 HyperGraphRAG 的差距尤其重要,因为这两种方法使用相同的底层超图表示;区别在于检索的证据是作为无序集合还是有序轨迹来处理。这突出了顺序意识在表示之外的贡献。



(a) Cross-horizon reasoning.

(b) Within-horizon factual retrieval

Figure 4: **Query-adaptive trajectories retrieved by OKH-RAG.** Upper layer: ordered hyperedges; lower layer: entities.

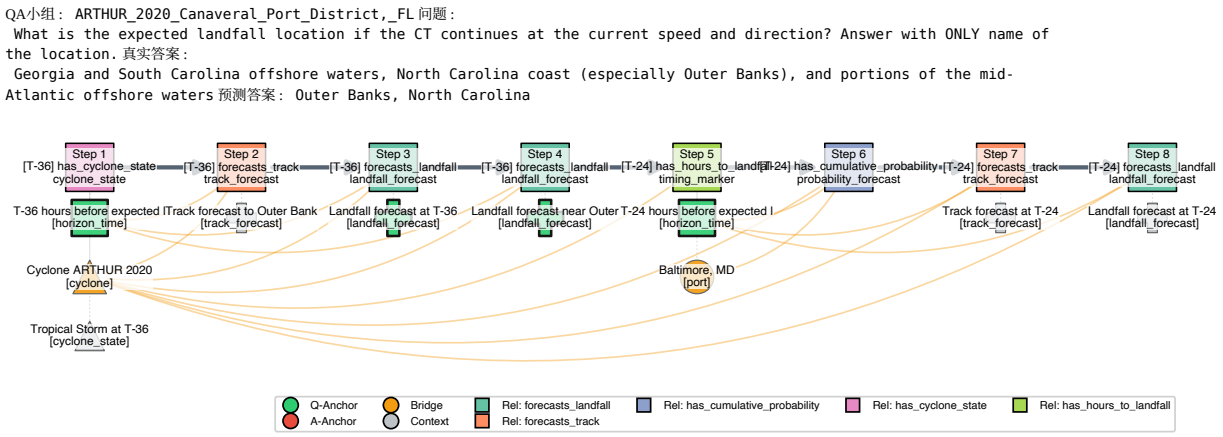
advantage of hypergraphs alone. The largest gains appear on MC and SA, suggesting that order-aware retrieval is especially beneficial when answers require multi-step disambiguation or synthesis across related evidence.

Table 1: Answer accuracy by question type. Best in **bold**; second-best underlined.

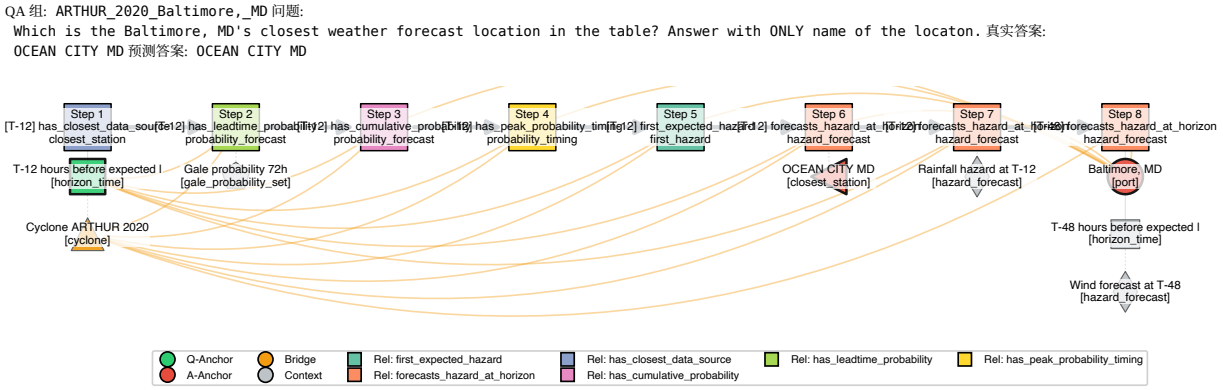
Method	TF	MC	SA	TD	Overall
Text-RAG	0.694	0.378	0.198	0.224	0.287
GraphRAG	0.806	0.506	0.321	0.362	0.414
HyperGraphRAG	0.819	0.620	0.435	0.432	0.511
OKH-RAG (ours)	**0.833**	**0.652**	**0.452**	**0.441**	**0.534**

4.5 Ablation Studies

Table 2 summarizes three ablations isolating the contribution of order-aware retrieval. The permutation stress test provides the most direct evidence: *OKH-RAG (shuffled)* reduces overall accuracy from 0.534 to 0.487, the largest drop observed, despite identical retrieved content. This shows that performance depends not only on which hyperedges are retrieved, but also on their order. The component analysis further indicates that all structural terms contribute positively. In particular, removing precedence consistency (μ) or phase coverage (ρ) yields the largest degradation, suggesting that effective reasoning requires both correct ordering and sufficient coverage. Removing entity continuity (ν) also reduces performance, while removing order coherence (λ) has a smaller effect, indicating its role is primarily to guide search toward better trajectories. The transition-model ablation reinforces this interpretation. Comparing *no order*, *heuristic order*, and *learned order* reveals a consistent hierarchy: performance improves as stronger forms of order modeling are introduced. Heuristic ordering provides modest gains over order-free retrieval, while the learned transition model performs best, capturing ordering patterns beyond fixed rules. Overall, these results show that (1) evidence order is



(a) 跨视野推理。



(b) 地平线内的事实检索

图 4:由 OKH-RAG 检索的查询自适应轨迹。上层:有序超边;下层:实体。

仅使用超图的优势。最大的收益出现在MC和SA上,这表明当答案需要跨相关证据进行多步骤消歧或综合时,考虑顺序的检索尤其有益。

表 1:一个 按问题类型的答案准确性。最佳的加粗;第二好的未 下划线的。

Method	TF	MC	SA	TD	Overall
Text-RAG	0.694	0.378	0.198	0.224	0.287
GraphRAG	0.806	0.506	0.321	0.362	0.414
HyperGraphRAG	0.819	0.620	0.435	0.432	0.511
OKH-RAG (ours)	**0.833**	**0.652**	**0.452**	**0.441**	**0.534**

4.5 消融研究

表2总结了三个消融实验,以隔离顺序感知检索的贡献。排列应力测试提供了最直接的证据:
OKH-RAG (shuffled) 将整体准确率从0.534降低到0.487,尽管检索到的内容相同,但仍是观察到的最大下降。这表明性能不仅取决于检索到哪些超边,还取决于它们的顺序。组件分析进一步表明,所有结构性项都具有正向贡献。特别是,移除优先顺序一致性(μ)或阶段覆盖(ρ)会导致性能下降最大,这表明有效的推理既需要正确的顺序,也需要足够的覆盖。移除实体连续性(ν)也会降低性能,而移除顺序一致性(λ)的影响较小,这表明其主要作用是引导搜索更好的轨迹。过渡模型的消融强化了这一解释。比较*no order*、*heuristic*、*order*和*learned order*显示了一个一致的层次结构:表现

a primary driver of reasoning quality, (2) effective trajectories must be both ordered and complete, and (3) learned precedence modeling provides additional benefits beyond heuristic constraints. Together, these findings confirm that OKH-RAG's gains arise specifically from order-aware trajectory retrieval rather than structured retrieval alone.

Table 2: **Ablation study of order-aware retrieval.**

Group	Variant	TF	MC	SA	TD	Overall	Δ
Full Model	**OKH-RAG (full)**	**0.833**	**0.652**	**0.452**	**0.441**	**0.534**	—
Shuffle	OKH-RAG (shuffled)	0.750	0.618	0.397	0.399	0.487	−0.047
Components	$-\ \lambda$ (order coherence)	0.833	0.650	0.449	0.445	0.532	−0.002
	$-\ \mu$ (precedence)	0.819	0.633	0.434	0.395	0.510	−0.024
	$-\ \nu$ (entity continuity)	0.819	0.646	0.440	0.407	0.519	−0.015
	$-\ \rho$ (phase coverage)	0.819	0.633	0.434	0.395	0.510	−0.024
Transition	No order	0.818	0.621	0.412	0.426	0.501	−0.033
	Heuristic order	0.821	0.625	0.426	0.430	0.509	−0.025

5 Conclusion

This work challenges a core assumption in retrieval-augmented generation: that retrieved evidence can be treated as an unordered set. We show that this assumption fails for a broad class of reasoning tasks where outcomes depend on how interactions unfold. To address this, we introduced OKH-RAG, which represents knowledge as order-aware hypergraphs and formulates retrieval as trajectory inference over higher-order interactions. Across experiments, OKH-RAG consistently outperforms permutation-invariant baselines, and ablations confirm that these gains arise specifically from modeling interaction order rather than from structured representation alone. These results highlight that retrieval quality depends not only on relevance, but also on organization: preserving the structure of evidence sequences is critical for faithful reasoning. More broadly, our findings suggest a shift from set-based to trajectory-based retrieval, enabling LLMs to reason over processes, dependencies, and evolving systems. This perspective is particularly important for domains such as scientific discovery, decision-making under uncertainty, and complex system analysis, where order is intrinsic to meaning.

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推理质量的主要驱动因素,(2) 有效的轨迹必须既有序又完整,(3) 学习的优先级建模提供了超出启发式约束的额外好处。这些发现共同证实,OKH-RAG 的提升具体来源于注重顺序的轨迹检索,而不仅仅是结构化检索。

表 2:顺序感知检索的消融实验。

Group	Variant	TF	MC	SA	TD	Overall	Δ
Full Model	**OKH-RAG (full)**	**0.833**	**0.652**	**0.452**	**0.441**	**0.534**	—
Shuffle	OKH-RAG (shuffled)	0.750	0.618	0.397	0.399	0.487	−0.047
Components	− λ (order coherence)	0.833	0.650	0.449	0.445	0.532	−0.002
	− μ (precedence)	0.819	0.633	0.434	0.395	0.510	−0.024
	− ν (entity continuity)	0.819	0.646	0.440	0.407	0.519	−0.015
	− ρ (phase coverage)	0.819	0.633	0.434	0.395	0.510	−0.024
Transition	No order	0.818	0.621	0.412	0.426	0.501	−0.033
	Heuristic order	0.821	0.625	0.426	0.430	0.509	−0.025

5 结论

这项工作挑战了增强检索生成(retrieval-augmented generation)中的一个核心假设:即检索到的证据可以被视为无序集合。我们展示了,对于一类广泛的推理任务来说,这一假设是行不通的,因为结果取决于交互的展开方式。为了解决这一问题,我们提出了 OKH-RAG,它将知识表示为有序超图,并将检索形式化为对高阶交互的轨迹推断。在各项实验中,OKH-RAG 始终优于置换不变基线,消融实验确认这些提升特别源于对交互顺序的建模,而不仅仅是结构化表示。这些结果强调,检索质量不仅取决于相关性,还取决于组织方式:保持证据序列的结构对于忠实推理至关重要。更广泛地说,我们的发现表明,应从基于集合的检索转向基于轨迹的检索,使大型语言模型能够对过程、依赖关系和演变进行推理。

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A Detailed Formulations of Methodology

This appendix provides the formal definitions, algorithmic details, and design rationale supporting the methodology in § 3. We follow the same three-stage organization: knowledge hypergraph construction (§ A.1–A.2), order learning (§ A.3–A.4), retrieval (§ A.5–A.7), and generation (§ A.8).

A.1 Formal definitions

The main text introduces entities and hyperedges as tuples. Here we expand on the design rationale behind each component and provide the complete relation vocabulary.

Entity. An entity $v \in \mathcal{V}$ is a tuple

$$v = (n_v,\ \tau_v,\ d_v,\ c_v), \tag{6}$$

where n_v is the entity name, grounded in the source hyperedge description ($n_v \subseteq s_e$); $\tau_v \in \mathcal{T}$ is a type drawn from a domain-specific type system; d_v is a natural-language explanation of the entity's role; and $c_v \in (0, 1]$ is a confidence score reflecting extraction certainty. The grounding constraint $n_v \subseteq s_e$ ensures traceability: every entity can be linked to a specific span in the source text, supporting provenance tracking and faithfulness evaluation.

The type system $\mathcal{T}$ distinguishes three functional categories, each serving a distinct role during retrieval:

- *Persistent objects* (PORT, CYCLONE): domain-level entities that persist across the scenario. These appear in hyperedges at many sequence positions and serve as *anchors*—their shared presence across steps is what entity continuity scoring (Ovlp) exploits to favor coherent trajectories.
- *Transient states* (CYCLONE_STATE, OPERATION_STATUS, HAZARD_FORECAST): horizon-specific instances capturing the system's configuration at a particular point. Each is typically associated with a single horizon, encoding how persistent objects evolve—e.g., a cyclone's category at T-96 versus T-48.
- *Temporal anchors* (HORIZON_TIME, e.g., `horizon:T-48`): structural entities that index interactions to positions in the precedence relation. Every horizon-grounded hyperedge includes exactly one temporal anchor, enabling horizon-scoped candidate selection and order enforcement during retrieval.

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方法论的详细制定

本附录提供了支持第3节方法论的正式定义、算法细节和设计原理。我们遵循相同的三阶段组织结构:知识超图构建(§ A.1–A.2)、顺序学习(§ A.3–A.4)、检索(§ A.5–A.7)和生成(§ A.8)。

A.1 正式定义

正文将实体和超边作为元组进行介绍。这里我们扩展每个组件背后的设计原理,并提供完整的关系词汇表。

实体。实体 $v \in \mathcal{V}$ 是一个元组

$$v = (n_v,\ \tau_v,\ d_v,\ c_v), \tag{6}$$

其中 n_v 是实体名称,基于源超边描述 ($n_v \subseteq s_e$);$\tau_v \in \mathcal{T}$ 是从特定领域类型系统中选取的类型;d_v 是对实体角色的自然语言解释;$c_v \in (0, 1]$ 是反映提取确定性的置信度评分。基础约束 $n_v \subseteq s_e$ 确保可追溯性:每个实体都可以关联到源文本中的特定片段,从而支持来源跟踪和可信度评估。

类型系统 $\mathcal{T}$ 区分三种功能类别,每种在检索过程中都扮演不同的角色:

- *Persistent objects* (PORT, CYCLONE):跨场景持久存在的域级实体。这些实体出现在许多序列位置的超边中,并作为*anchors*——它们在各步骤间的共享存在是实体连续性评分(Ovlp))利用以偏好连贯轨迹的依据。
- *Transient states* (CYCLONE_STATE,OPERATION_STATUS,HAZARD_FORECAST):特定时间点捕获系统配置的按时间水平划分的实例。每个实例通常与单一时间水平相关,编码持久对象如何演变——例如,飓风在T-96与T-48的等级。
- *Temporal anchors* (HORIZON_TIME,例如,horizon:T-48):索引交互到优先关系中位置的结构实体。每个以地平线为基础的超边都包括恰好一个时间锚点,使得在检索过程中能够进行地平线范围内的候选选择和顺序执行。

This typology directly governs how reasoning chains are traced: persistent objects provide cross-horizon continuity, transient states provide within-horizon specificity, and temporal anchors provide the positional scaffolding that makes order-aware retrieval possible.

Hyperedge. A hyperedge $e \in \mathcal{E}$ is a tuple

$$e = (V_e,\ r_e,\ s_e,\ \mathbf{a}_e,\ c_e), \tag{7}$$

with five components:

- $V_e = \{v_1, \ldots, v_m\} \subseteq \mathcal{V}$: the participating entity set ($m \geq 2$). The higher-order cardinality of V_e is what distinguishes hyperedges from binary edges and enables atomic representation of multi-factor interactions.
- $r_e \in \mathcal{R}$: a typed relation from the controlled vocabulary (Table 3), categorizing the *kind* of interaction and enabling phase-aware retrieval.
- s_e: a natural language description preserving the full semantic content of the interaction. Unlike structured triples, s_e can express complex conditional dependencies involving multiple entities and quantitative thresholds simultaneously, and is directly used to compute the hyperedge embedding $\mathbf{h}_e$.
- $\mathbf{a}_e = \{(k_j, u_j)\}_{j=1}^{A}$: key–value attributes recording quantitative and categorical properties, enabling precise factual grounding during generation and numeric question answering.
- $c_e \in (0, 1]$: extraction confidence, available for downstream retrieval weighting.

Relation vocabulary. The vocabulary $\mathcal{R}$ spans the complete life cycle of domain processes, from initial state characterization through impact and recovery (Table 3). The ordering of families 1–12 corresponds to the canonical phase progression used for within-horizon precedence (§ A.3); family 13 operates between horizons. All relation strings are normalized to this vocabulary via an alias mapping that absorbs variation in LLM-generated labels (e.g., `closes_port` → `has_operation_status`).

Table 3: Relation vocabulary $\mathcal{R}$: 13 semantic families in canonical phase order.

#	Family	Representative relations
1	Cyclone state	`has_cyclone_state`, `has_category_state`, `has_motion`
2	Track & landfall	`forecasts_track`, `forecasts_landfall`
3	Timing	`has_hours_to_landfall`, `has_forecast_window`
4	Advisory	`has_watch_status`, `has_warning_status`
5	Probability	`has_leadtime_probability`, `has_cumulative_probability`
6	Hazard forecast	`forecasts_hazard_at_horizon`
7	Hazard observation	`observes_hazard_at_horizon`
8	Threshold	`has_threshold_status`
9	Additional hazards	`has_additional_hazard`
10	Operations	`has_operation_status`, `affects_vessel_handling`
11	Impact	`has_impact_prediction`, `causes_operational_disruption`
12	Recovery	`has_recovery_status`, `starts_recovery`
13	Cross-horizon	`forecast_updates_to`, `intensifies_to`, `changes_status_to`

A.2 N-ary relational extraction

The extraction pipeline converts unstructured documents into the knowledge hypergraph $\mathcal{H}$. Three aspects of the pipeline merit detailed description: the core extraction step, the per-horizon decomposition strategy, and the post-processing normalization.

Core extraction. For each document $d \in \mathcal{K}$, a language model π (GPT-4o) receives an extraction prompt p_{ext} and produces n-ary relational facts:

$$\mathcal{F}^d = \{f_1, \ldots, f_k\} \sim \pi(\mathcal{F} \mid p_{\text{ext}},\ d), \tag{8}$$

where each $f_i = (e_i, V_{e_i})$ pairs a hyperedge with its entity set. The prompt instructs the model to segment the input into knowledge fragments (yielding s_e), recognize and type all entities (yielding V_e with τ_v and d_v), and assign relation labels from $\mathcal{R}$ with quantitative attributes. Output is constrained to a strict JSON schema matching the definitions above.

这种类型学直接决定了推理链条的追踪方式:持久对象提供跨视野的连续性,瞬态状态提供视野内的特定性,时间锚点提供位置支撑,使得有序感知的检索成为可能。

超边。一个超边 $e \in \mathcal{E}$ 是一个元组

$$e = (V_e,\ r_e,\ s_e,\ \mathbf{a}_e,\ c_e), \tag{7}$$

有五个组成部分:

- $V_e = \{v_1, \ldots, v_m\} \subseteq \mathcal{V}$:参与实体集($m \geq 2$)。$V_e$ 的高阶基数是区分超边和二元边的关键,并且使多因素交互的原子表示成为可能。
- $r_e \in \mathcal{R}$:来自受控词汇表(表3)的一个类型化关系,对交互的*kind*进行分类,并支持阶段感知的检索。
- s_e:一种自然语言描述,保留了交互的全部语义内容。与结构化三元组不同,s_e 可以同时表达涉及多个实体和定量阈值的复杂条件依赖,并直接用于计算超边嵌入 $\mathbf{h}_e$。
- $\mathbf{a}_e = \{(k_j, u_j)\}_{j=1}^{A}$:键-值属性记录定量和分类属性,使生成过程和数值问题回答中能够实现精确的事实基础。
- $c_e \in (0, 1]$:提取置信度,可用于下游检索加权。

关系词汇。词汇 $\mathcal{R}$ 覆盖了领域过程的完整生命周期,从初始状态描述到影响和恢复(表 3))。家族 1–12 的排序对应于地平线内优先级使用的规范阶段进展(§ A.3);家族 13 在地平线之间运行。所有关系字符串都通过别名映射归一化到此词汇,以吸收 LLM 生成标签的变体(例如,closes_port → has_operation_status)。

表 3:关系词汇 $\mathcal{R}$:规范阶段顺序的 13 个语义家族。

#	Family	Representative relations
1	Cyclone state	`has_cyclone_state`, `has_category_state`, `has_motion`
2	Track & landfall	`forecasts_track`, `forecasts_landfall`
3	Timing	`has_hours_to_landfall`, `has_forecast_window`
4	Advisory	`has_watch_status`, `has_warning_status`
5	Probability	`has_leadtime_probability`, `has_cumulative_probability`
6	Hazard forecast	`forecasts_hazard_at_horizon`
7	Hazard observation	`observes_hazard_at_horizon`
8	Threshold	`has_threshold_status`
9	Additional hazards	`has_additional_hazard`
10	Operations	`has_operation_status`, `affects_vessel_handling`
11	Impact	`has_impact_prediction`, `causes_operational_disruption`
12	Recovery	`has_recovery_status`, `starts_recovery`
13	Cross-horizon	`forecast_updates_to`, `intensifies_to`, `changes_status_to`

A.2 N -元关系抽取

提取管道将非结构化文档转换为知识超图 $\mathcal{H}$。该管道的三个方面值得详细描述:核心提取步骤、逐视角分解策略以及后处理归一化。

核心提取。对于每个文档 $d \in \mathcal{K}$,语言模型 π (GPT-4o) 接收一个提取提示 p_{ext} 并生成 n 元关系事实:

$$\mathcal{F}^d = \{f_1, \ldots, f_k\} \sim \pi(\mathcal{F} \mid p_{\text{ext}},\ d), \tag{8}$$

其中每个 $f_i = (e_i, V_{e_i})$ 将一个超边与其实体集配对。提示指示模型将输入分割为知识片段(生成 s_e)、识别并标注所有实体类型(生成 V_e,包含 τ_v 和 d_v),并从 $\mathcal{R}$ 分配带有量化属性的关系标签。输出受限于严格的 JSON 模式,与上述定义相匹配。

Per-horizon extraction. Multi-horizon scenario documents can span thousands of tokens. Single-call extraction risks truncation, entity conflation across horizons, and reduced attribute coverage. We therefore split each document at horizon headers (detected via regex patterns such as "T-48 hours before expected landfall:") and extract each block independently. This per-horizon strategy yields substantially higher entity and attribute coverage, at the cost of a subsequent merge step.

Post-processing. Five normalization steps ensure consistency across per-horizon extractions:

1. *Relation normalization*: relation strings are mapped to $\mathcal{R}$ via the alias table and fuzzy matching, absorbing LLM-generated variation.
2. *Entity ID canonicalization*: identifiers are rewritten to a hierarchical convention encoding family, storm, port, and horizon (e.g., `wind_fcst:IRMA:port_arthur:T-48`), ensuring cross-block consistency.
3. *Horizon entity injection*: a canonical temporal anchor (`horizon:T-48`) is created for each detected horizon and added to all hyperedges at that horizon, making horizon membership structurally explicit.
4. *Cross-horizon edge synthesis*: for entity families at multiple horizons, synthetic hyperedges (`forecast_updates_to`, `changes_probability_to`) are created to represent evolution, providing the cross-horizon links that the precedence construction requires.
5. *Deduplication*: unique hyperedge IDs are computed as hashes of relation, entity set, and evidence text; exact duplicates are collapsed.

Aggregation yields the complete hypergraph:

$$\mathcal{H} = \Big(\bigcup_{d \in \mathcal{K}} \bigcup_{f_i \in \mathcal{F}^d} V_{e_i}, \quad \bigcup_{d \in \mathcal{K}} \{e_i \mid f_i \in \mathcal{F}^d\} \Big). \tag{9}$$

A.3 Precedence construction

The precedence relation $\prec$ over $\mathcal{E}$ is constructed from domain-informed structural rules that produce a DAG within each knowledge group.

Sequence-indexed representation. Hyperedges are assigned to sequence positions $\ell \in \{1, \ldots, L\}$, yielding states $\mathcal{H}^{(\ell)} = (\mathcal{V}, \mathcal{E}^{(\ell)})$ and the precedence relation:

$$e_i \prec e_j \iff \exists\, \ell_1 < \ell_2 \text{ s.t. } e_i \in \mathcal{E}^{(\ell_1)},\ e_j \in \mathcal{E}^{(\ell_2)}. \tag{10}$$

Structural rules. Four rule families determine $\prec$:

Within-horizon phase ordering. At the same horizon, hyperedges are ordered by semantic phase following Table 3: cyclone state → track → landfall → timing → advisory → probability → hazard → threshold → operation → impact → recovery. This reflects logical dependency: an operational decision presupposes a hazard assessment, which presupposes an advisory.

Cross-horizon family evolution. Hyperedges in the same semantic family at different horizons follow decreasing horizon order: T-96 $\prec$ T-72 $\prec$ T-48 $\prec$ T-24 $\prec$ T-12. This ensures that evolving information is presented chronologically.

Causal chain rules. Four within-horizon inter-phase constraints encode dominant causal pathways: advisory $\prec$ hazard forecasts, hazard assessments $\prec$ operational decisions, hazard assessments $\prec$ impact predictions, and impact predictions $\prec$ recovery status.

Family-to-change ordering. Within-horizon hyperedges in a semantic family precede any cross-horizon change hyperedges (family 13) for the same family, ensuring the "before" state is presented before the transition connecting it to the next horizon.

Canonical trajectory. Topological sorting the resulting DAG with deterministic tie-breaking (phase rank, family rank, lead time, text position) yields a canonical linear ordering per knowledge group. This ordering serves two purposes: it provides the ground-truth trajectory for document-order supervision in training (§ A.4), and it defines the structural precedence that $\mathrm{Prec}(\gamma)$ enforces during retrieval.

按地平线提取。多地平线场景文档可能包含数千个标记。一次性提取存在截断风险、跨地平线实体混淆以及属性覆盖率降低的问题。因此,我们在地平线标题处(通过正则表达式模式检测,如“预期登陆前 T-48 小时:”)拆分每个文档,并独立提取每个块。这种按地平线的策略可以显著提高实体和属性的覆盖率,但需要后续的合并步骤。

后处理。五个归一化步骤确保各时段提取的一致性:

1. *Relation normalization*:关系字符串通过别名表和模糊匹配映射到 $\mathcal{R}$,吸收由大语言模型生成的变体。 2. *Entity ID canonicalization*:标识符被重写为分层约定,编码族、风暴、港口和时间地平线(例如,wind_fcst:IRMA:port_arthur:T-48),确保跨区块的一致性。 3. *Horizon entity injection*:为每个检测到的时间地平线创建一个标准的时间锚(horizon:T-48),并将其添加到该时间地平线的所有超边中,使地平线成员关系在结构上明确。 4. *Cross-horizon edge synthesis*:对于存在于多个时间地平线的实体族,创建合成超边(forecast_updates_to, changes_probability_to)以表示演变,提供前置构建所需的跨地平线链接。

5. *Deduplication*:唯一的超边ID是通过关系、实体集合和证据文本的哈希计算得到的;完全重复的将被合并。

聚合产生完整的超图:

$$\mathcal{H} = \Big(\bigcup_{d \in \mathcal{K}} \bigcup_{f_i \in \mathcal{F}^d} V_{e_i},\ \bigcup_{d \in \mathcal{K}} \{ e_i \mid f_i \in \mathcal{F}^d \} \Big). \tag{9}$$

A.3 优先结构

在每个知识组内,由领域信息指导的结构规则构建出的有向无环图中,$\prec$ 对 $\mathcal{E}$ 的优先关系被建立。

序列索引表示。超边被分配到序列位置 $\ell \in \{1, \ldots, L\}$,产生状态 $\mathcal{H}^{(\ell)} = (\mathcal{V}, \mathcal{E}^{(\ell)})$ 和优先关系:

$$e_i \prec e_j \iff \exists\, \ell_1 < \ell_2 \text{ s.t. } e_i \in \mathcal{E}^{(\ell_1)},\ e_j \in \mathcal{E}^{(\ell_2)}. \tag{10}$$

结构规则。四类规则决定 $\prec$:

Within-horizon phase ordering. 在同一水平线上,超边按照语义阶段排序,如表3所示:气旋状态 $\rightarrow$ 轨迹 $\rightarrow$ 登陆 $\rightarrow$ 时间 $\rightarrow$ 咨询 $\rightarrow$ 概率 $\rightarrow$ 危险 $\rightarrow$ 阈值 $\rightarrow$ 操作 $\rightarrow$ 影响 $\rightarrow$ 恢复。这反映了逻辑依赖关系:操作决策以危险评估为前提,危险评估以咨询为前提。

Cross-horizon family evolution. 属于同一语义类别的超边在不同时间层次上遵循递减的时间层次顺序:T-96 $\prec$ T-72 $\prec$ T-48 $\prec$ T-24 $\prec$ T-12。这确保了演变信息按时间顺序呈现。

Causal chain rules. 四个视域内相位约束编码了主要的因果路径:咨询 $\prec$ 危害预测、危害评估 $\prec$ 操作决策、危害评估 $\prec$ 影响预测,以及影响预测 $\prec$ 恢复状态。

Family-to-change ordering. 在语义族中的地平线内超边先于同一族的任何跨地平线变化超边(族13),以确保“之前”状态在连接到下一个地平线的转换之前呈现。

规范轨迹。对生成的有向无环图(DAG)进行拓扑排序,并使用确定性平局打破策略(阶段等级、家族等级、提前时间、文本位置),可以为每个知识组得到规范的线性排序。此排序有两个目的:一是为训练中的文档顺序监督提供真实轨迹(§ A.4),二是定义 Prec(γ) 在检索过程中强制执行的结构优先顺序。

A.4 Transition model training

The learned transition model P_θ complements the structural precedence $\prec$ by capturing soft ordering preferences that rules alone cannot express—e.g., that a wind forecast is more likely to be followed by a surge forecast than by a recovery update, even when both are valid under $\prec$.

Embedding. Hyperedges are embedded as $\mathbf{h}_e \in \mathbb{R}^{1536}$ via OpenAI `text-embedding-3-small`. The input text concatenates the relation type, evidence string s_e, entity names and types, and key attribute values, providing both structural and semantic information.

Low-rank parameterization. The weight matrix $W = U^\top V$ with $U, V \in \mathbb{R}^{64\times 1536}$ yields $196{,}608$ parameters versus $\sim$2.4M for the full matrix, while preserving the asymmetry essential for directional modeling. The partition function is approximated via sampled softmax with $K = 64$ global negatives per example.

Self-supervised signals. Three signal families, each exploiting a different form of implicit order, compose the training set:

Document order ($\mathcal{P}_{\text{doc}}$). For co-occurring hyperedges (e_i, e_j) with $\pi(e_i) < \pi(e_j)$, the pair joins $\mathcal{P}$; the reversal (e_j, e_i) and random cross-group pairs join $\mathcal{N}$. This is the most abundant but noisiest signal: document order is a proxy for, not a guarantee of, logical precedence.

Entity-overlap consistency ($\mathcal{P}_{\text{ent}}$). Pairs sharing entities ($|V_{e_i} \cap V_{e_j}| \geq 1$) within the same knowledge group join $\mathcal{P}$ if they appear in canonical trajectory order (§ A.3). This signal is sparser but higher-quality: shared entities imply topical relatedness, and the canonical order provides a principled direction.

Retrieval-induced preference ($\mathcal{P}_{\text{ret}}$). After initial training, retrieval traces from correct answers yield additional positive consecutive pairs, creating a self-training loop: better retrievals improve the model, which improves subsequent retrievals.

A.5 Order sensitivity: proof and discussion

Proposition 1 asserts that permutation-invariant retrieval is insufficient for order-sensitive reasoning. We provide a constructive proof and discuss scope.

Constructive proof. Let k_1, k_2, k_3 be three interactions:

k_1 : At T-96, the storm is a tropical depression (Category 0).

k_2 : At T-48, the storm has intensified to Category 2; uncertainty cone covers the port.

k_3 : At T-12, gale-force wind probability exceeds 80%; port restricts vessel movements.

For the query "Was the port restriction justified?", the sequence (k_1, k_2, k_3) presents progressive escalation supporting "yes," while (k_3, k_2, k_1) presents the restriction before its justification, supporting "premature." Since these yield different answer distributions, $P(y \mid k_1, k_2, k_3, q) \neq P(y \mid k_3, k_2, k_1, q)$. □

Scope. The proposition requires only that *some* query's answer depends on order—not that all queries do. This minimal requirement is satisfied in any domain with precedence-dependent outcomes: clinical diagnosis, legal reasoning, cascading engineering failures, or operational decision-making under evolving hazards.

A.6 Retrieval scoring terms

All terms in Equation 5 are designed for efficient incremental computation during beam search.

Relevance. $\text{Rel}(e, q) = \cos(\mathbf{h}_e, \mathbf{q})$: cosine similarity in $\mathbb{R}^{1536}$ between hyperedge and query embeddings. Precomputed once per query for all candidates.

Order coherence. $\text{Trans}(\gamma) = \sum_{k=1}^{L-1} \log P_\theta(e^{(k+1)} \mid e^{(k)})$: cumulative log-transition probability, computed incrementally from the precomputed $|\mathcal{C}| \times |\mathcal{C}|$ matrix.

A.4 转换模型训练

学习到的迁移模型 P_θ 通过捕捉仅靠规则无法表达的软排序偏好,补充了结构优先性 $\prec$——例如,即使在 $\prec$ 下两者都是有效的,风力预报更可能被海潮突增预报跟随,而不是由恢复更新跟随。

嵌入。超边通过 OpenAI text-embedding-3-small 被嵌入为 $\mathbf{h}_e \in \mathbb{R}^{1536}$。输入文本将关系类型、证据字符串 s_e 、实体名称和类型以及关键属性值连接在一起,同时提供结构和语义信息。

低秩参数化。权重矩阵 $W = U^\top V$ 与 $U, V \in \mathbb{R}^{64\times 1536}$ 产生 196,608 个参数,而完整矩阵则有 ∼2.4M 个参数,同时保留了对方向建模至关重要的非对称性。分区函数通过每个示例 $K =$ 个 64 个全局负样本的采样 softmax 进行近似。

自监督信号。三个信号家族,每个家族利用一种不同形式的隐含顺序,组成训练集:

Document order ($\mathcal{P}_{\text{doc}}$). 对于与 $\pi(e_i) < \pi(e_j)$ 共同出现的超边 (e_i, e_j),该对加入 $\mathcal{P}$;反向 (e_j, e_i) 和随机跨组对加入 $\mathcal{N}$。这是最丰富但最嘈杂的信号:文档顺序是逻辑先后顺序的代理,而不是保证。

Entity-overlap consistency ($\mathcal{P}_{\text{ent}}$). 在相同知识组内共享实体($|V_{e_i} \cap V_{e_j}| \geq 1$)的对,如果它们出现在规范轨迹顺序中 (§ A.3),则加入 $\mathcal{P}$。这种信号更稀疏但质量更高:共享实体意味着主题相关性,而规范顺序提供了一个原则性的方向。

Retrieval-induced preference ($\mathcal{P}_{\text{ret}}$). 初始训练后,来自正确答案的检索痕迹会产生额外的正向连续对,形成自我训练循环:更好的检索提升模型,从而改善后续的检索。

A.5 顺序敏感性:证明与讨论

命题 1 断言,置换不变的检索对于顺序敏感的推理是不足够的。我们提供了一个建设性的证明并讨论了其范围。

建设性证明。设 k_1, k_2, k_3 为三个相互作用:

k_1:在T-96时,风暴是热带低压(0级)。k_2:在T-48时,风暴已增强至2级;不确定性范围覆盖港口。k_3:在T-12时,强风概率超过80%;港口限制船舶通行。

对于查询“港口限制是否合理?”,序列(k_1, k_2, k_3)呈现了支持“是”的逐步升级,而(k_3, k_2, k_1)在其理由之前呈现了限制,支持“过早”。由于这些产生了不同的答案分布,$P(y \mid k_1, k_2, k_3, q) \neq P(y \mid k_3, k_2, k_1, q)$。 □

范围。该命题仅要求 *some* 查询的答案依赖于顺序——并不要求所有查询都如此。这个最小要求在任何具有先后依赖结果的领域都能得到满足:临床诊断、法律推理、连锁工程故障,或在变化的危险下进行的操作决策。

A.6 检索评分术语

方程5中的所有项都旨在在束搜索过程中进行高效的增量计算。

相关性。Rel$(e, q) = \cos(\mathbf{h}_e, \mathbf{q})$:$\mathbb{R}^{1536}$中超边与查询嵌入之间的余弦相似度。每个查询对所有候选项预先计算一次。

顺序一致性。Trans$(\gamma) = \sum_{k=1}^{L-1}$ 日志 $P_\theta(e^{(k+1)} \mid e^{(k)})$:累积日志转移概率,从预先计算的 $|\mathcal{C}| \times |\mathcal{C}|$ 矩阵中逐步计算。

Precedence consistency.

$$\mathrm{Prec}(\gamma) = \frac{\left|\{k : e^{(k)} \prec e^{(k+1)}\}\right|}{\left|\{k : (e^{(k)}, e^{(k+1)}) \in (\prec \cup \succ)\}\right|}. \tag{11}$$

Fraction of ordered consecutive pairs consistent with $\prec$; unrelated pairs are excluded from the denominator.

Entity continuity.

$$\mathrm{Ovlp}(\gamma) = \frac{1}{L-1} \sum_{k=1}^{L-1} \frac{|V_{e^{(k)}} \cap V_{e^{(k+1)}}|}{|V_{e^{(k)}} \cup V_{e^{(k+1)}}|}. \tag{12}$$

Mean Jaccard similarity of entity sets between consecutive steps.

Phase coverage.

$$\mathrm{Cov}(\gamma) = \frac{\left|\{\phi(e^{(k)}) : k = 1, \ldots, L\} \cap \mathcal{S}\right|}{|\mathcal{S}|}, \tag{13}$$

where $\phi(e)$ maps a hyperedge to its reasoning phase and $\mathcal{S} = \{\texttt{advisory}, \texttt{hazard_forecast}, \texttt{hazard_observation}, \texttt{operation_stat}$

A.7 Inference algorithms

Beam search. Algorithm 1 formalises order-aware beam search. The key distinction from standard beam search is *sequence-dependent scoring*: the score of appending e' to beam β depends on the last element (via P_θ), the accumulated entity set (via Ovlp), and the full prefix (via the diversity penalty). This dependency is what transforms retrieval from independent ranking into trajectory inference.

Algorithm 1 Order-Aware Beam Search

Require: Query q, candidates $\mathcal{C}$, model P_θ, width B, length L, weights λ, μ, ν, ρ
Precompute $\mathrm{Rel}(e, q)$ and $\log P_\theta(e_j \mid e_i)$ for all $e, e_i, e_j \in \mathcal{C}$
$\mathcal{B} \leftarrow$ top-$2B$ singletons by $\mathrm{Rel}(e, q)$
for $\ell = 2, \ldots, L$ **do**
  $\mathcal{B}' \leftarrow \emptyset$
  for each $\beta = (e^{(1)}, \ldots, e^{(\ell-1)}) \in \mathcal{B}$, each $e' \in \mathcal{C} \setminus \beta$ **do**
    $s \leftarrow \mathrm{Rel}(e', q) + \lambda \log P_\theta(e' \mid e^{(\ell-1)}) + \mu \cdot \mathbb{1}[e^{(\ell-1)} \prec e']$
    $s \leftarrow s + \nu \cdot J(V_{e^{(\ell-1)}}, V_{e'}) + \rho \cdot \Delta\mathrm{Cov}(e', \beta)$
    $\mathcal{B}' \leftarrow \mathcal{B}' \cup \{(\beta \oplus e',\ \mathrm{score}(\beta) + s)\}$
  end for
  Apply diversity penalty; retain top-B by score
end for
return Top-N trajectories with score breakdowns

Viterbi dynamic programming. For exact optimisation under the two-term objective ($\mathrm{Rel} + \lambda \cdot \mathrm{Trans}$):

$$\mathrm{dp}[0][j] = \mathrm{Rel}(e_j, q), \tag{14}$$

$$\mathrm{dp}[\ell][j] = \max_i \big\{\mathrm{dp}[\ell-1][i] + \mathrm{Rel}(e_j, q) + \lambda \log P_\theta(e_j \mid e_i)\big\}, \tag{15}$$

with backpointers for trajectory recovery. Complexity: $O(L \cdot |\mathcal{C}|^2)$. Viterbi optimizes the two-term objective exactly but omits auxiliary terms; it serves primarily as a quality ceiling for beam search.

Candidate scoping. Two stages reduce the search space: (1) top-K (K=80) cosine retrieval from the global index; (2) group-aware expansion via group membership and one-hop entity overlap, with 40% of slots reserved for the query's own group when known. The pool is capped at $|\mathcal{C}| = 150$.

A.8 Generation details

Evidence format. Each hyperedge $e^{(k)}$ in a trajectory is presented as:

优先顺序一致性。

$$\mathrm{Prec}(\gamma)=\frac{\left|\{k:e^{(k)}\prec e^{(k+1)}\}\right|}{\left|\{k:(e^{(k)},e^{(k+1)})\in(\prec\cup\succ)\}\right|}. \tag{11}$$

与 $\prec$ 一致的有序连续对的分数;无关对不计入分母。

实体连续性。

$$\mathrm{Ovlp}(\gamma)=\frac{1}{L-1}\sum_{k=1}^{L-1}\frac{|V_{e^{(k)}}\cap V_{e^{(k+1)}}|}{|V_{e^{(k)}}\cup V_{e^{(k+1)}}|}. \tag{12}$$

连续步骤之间实体集合的平均Jaccard相似度 附言

相位覆盖

$$\mathrm{Cov}(\gamma)=\frac{\left|\{\phi(e^{(k)}):k=1,\dots,L\}\cap\mathcal{S}\right|}{|\mathcal{S}|}, \tag{13}$$

在哪里 $\phi(e)$ 将超边映射到其推理阶段和 $\mathcal{S}=\{$咨询, 危险预报, 危险观察, 操作_s tatu

A.7 推理算法

束搜索。算法 1 形式化了考虑顺序的束搜索。与标准束搜索的关键区别在于 *sequence-dependent scoring*:将 e' 添加到束 β 的得分取决于最后一个元素(通过 P_θ)、累积实体集(通过 Ovlp)以及完整的前缀(通过多样性惩罚)。这种依赖性将检索从独立排序转变为轨迹推断。

算法 1 顺序感知束搜索

要求:查询 q,候选 $\mathcal{C}$,模型 P_θ,宽度 B,长度 L,权重 λ,μ,ν,ρ 1:预计算所有 $e,e_i,e_j\in\mathcal{C}$ 的 Rel(e,q) 和对数 $P_\theta(e_j\mid e_i)$ 2:按 Rel(e,q) 选择前-$2B$ 单元素 $\mathcal{B}\leftarrow$ 3:对于 $\ell=2,\dots,L$ 执行 4:$\mathcal{B}'\leftarrow\emptyset$ 5:对于每个 $\beta=(e^{(1)},\dots,e^{(\ell-1)})\in\mathcal{B}$,每个 $e'\in\mathcal{C}\setminus\beta$ 执行 6:$s\leftarrow$ Rel$(e',q)+\lambda$ 对数 $P_\theta(e'\mid e^{(\ell-1)})+\mu\cdot\mathbb{1}[e^{(\ell-1)}\prec e']$ 7:$s\leftarrow s+\nu\cdot J(V_{e^{(\ell-1)}},V_{e'})+\rho\cdot\Delta$Cov$(e',\beta)$ 8:$\mathcal{B}'\leftarrow\mathcal{B}'\cup\{(\beta\oplus e',$ 分数$(\beta)+s)\}$ 9:结束循环 10:应用多样性惩罚;按分数保留前-B 11:结束循环 12:返回带有分数细分的前-N 条轨迹

Viterbi 动态规划。在两项目标(Rel $+\lambda\cdot$ Trans)下的精确优化:

$$\mathrm{dp}[0][j]=\mathrm{Rel}(e_j,q), \tag{14}$$

$$\mathrm{dp}[\ell][j]=\max_i\big\{\mathrm{dp}[\ell-1][i]+\mathrm{Rel}(e_j,q)+\lambda\log P_\theta(e_j\mid e_i)\big\}, \tag{15}$$

带有回指以恢复轨迹。复杂度:$O(L\cdot|\mathcal{C}|^2)$。Viterbi 精确优化双项目标,但省略辅助项;它主要作为束搜索的质量上限。

候选范围确定。两个阶段减少搜索空间:(1) 从全局索引中进行前 K (K=80) 余弦检索;(2) 通过组成员身份和一跳实体重叠进行基于组的扩展,当已知查询所属组时,为查询自身的组保留 40% 的位置。候选池上限为 $|\mathcal{C}|=150$。

A.8 生成详情

证据格式。轨迹中的每个超边 $e^{(k)}$ 表示为:

```
[Step k] [T-XX] [phase=...] [family=...]
  Relation: r_e
  Evidence: s_e
  Reasoning: within_horizon, hazard_to_operation, ...
  Entities: entity1 [type]; entity2 [type]; ...
```

Step indices provide ordering cues; horizon and phase labels provide temporal and logical context; reasoning tags indicate causal pathway participation; entity lists enable cross-step tracking. A trajectory-level quality summary is prepended when available.

Multi-trajectory synthesis. The *single-call* mode presents all N trajectories as numbered reasoning paths in one prompt. The model is instructed to read all paths before answering, treat convergence as a reliability signal, and note which trajectory supports the final answer. The *fallback* mode generates per-trajectory answers and aggregates via confidence-weighted voting (categorical) or averaging (numeric). The single-call mode is preferred: it enables direct cross-referencing rather than post-hoc reconciliation.

Output format. Structured JSON with fields `answer`, `confidence` $\in [0, 1]$, and `rationale`. Type-specific constraints ensure evaluation-friendly output: "Yes"/"No" for true/false, letter labels for multiple choice, numeric values for quantitative questions, free text for descriptions.

[步骤 k] [T-XX] [阶段=...] [系列=...] 关系: r_e 证据: s_e 推理: 在
视野内,操作危险,... 实体: entity1 [类型]; entity2 [类型]; ...

步骤索引提供排序提示;地平线和阶段标签提供时间和逻辑背景;推理标签指示因果路径参与;实体列表支持跨步骤跟踪。当可用时,会在前面附加轨迹级别的质量总结。

多轨迹综合。*single-call* 模式将所有 N 轨迹作为编号推理路径呈现在一个提示中。模型被指示在回答前阅读所有路径,将收敛视为可靠性信号,并注明哪条轨迹支持最终答案。*fallback* 模式生成每条轨迹的答案,并通过置信度加权投票(分类)或平均(数值)进行汇总。单次调用模式更受推荐:它能够直接进行交叉引用,而非事后调和。

输出格式。结构化 JSON,包含字段 answer、confidence $\in [0, 1]$ 和 rationale。类型特定约束确保便于评估的输出:对真/假使用“是”/“否”,多项选择使用字母标签,定量问题使用数值,描述性问题使用自由文本。